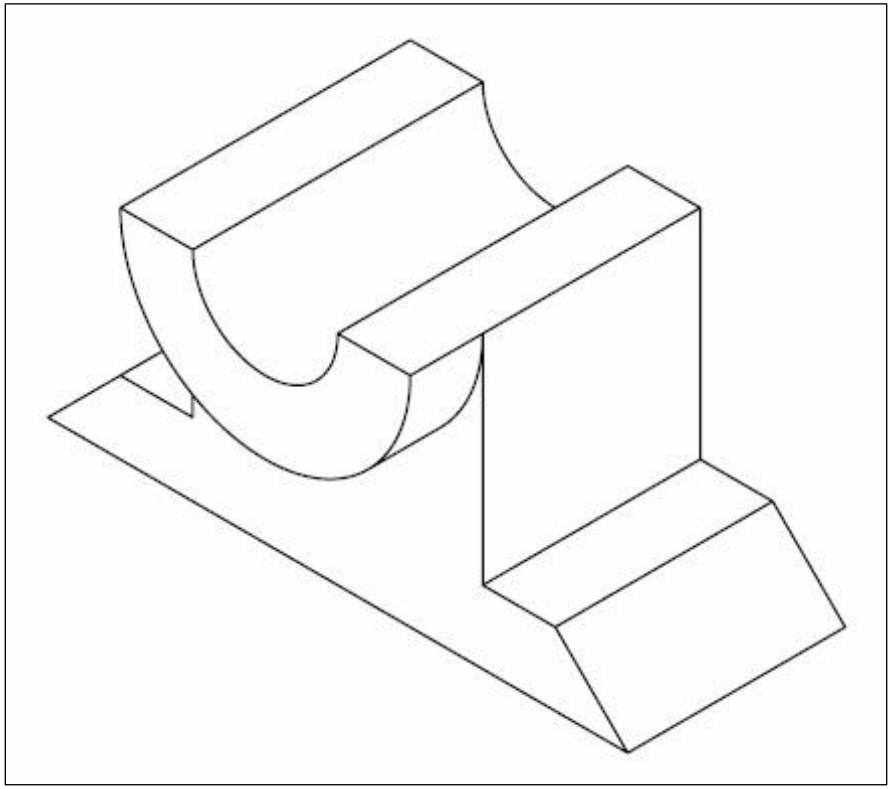
SolidWorks Education

Detailed Drawing Exercises



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Autodesk Inventor 2025

Student Modeling Workflow - SW Education Lessons 1-50

How to use this packet

- Each Inventor workflow page appears immediately before its matching SolidWorks Education drawing.
- All dimensions are in millimeters (mm). Use the dimensions on the drawing; do not estimate dimensions from the image.
- The listed Inventor 2025 commands are suggested modeling strategies. More than one correct feature sequence may exist.
- Fully constrain primary sketches whenever practical and use geometric constraints before adding unnecessary dimensions.
- After modeling, verify Front, Top, Right, and Isometric views against the supplied drawing.
- Recommended file naming: SW_Lesson_01.ipt through SW_Lesson_50.ipt.

Certification practice focus

Sketch creation, geometric constraints, dimensional constraints, Extrude, Extrude Cut, Hole, Fillet, Chamfer, Rib, work geometry, feature sequencing, model inspection, and orthographic visualization.

AUTODESK INVENTOR 2025 - LESSON 01

Saddle Support

SUGGESTED COMMANDS

Start 2D Sketch | Line | Centerline | Arc | General Dimension | Tangent | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Start a new Standard (mm).ipt part.
2. Sketch the main front profile using Line and the dimensions shown.
3. Add the 45 degree end geometry.
4. Establish a vertical Centerline through the saddle.
5. Create the curved saddle using an Arc and dimension the inner radius R10.
6. Apply Horizontal, Vertical, Coincident, Tangent, and dimensional constraints.
7. Extrude the completed profile to the depth shown in the right-side view.
8. Inspect the model in Isometric View and compare all four views.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Fully constrained sketches, arcs, angles, extrusion.

MODEL REQUIREMENTS

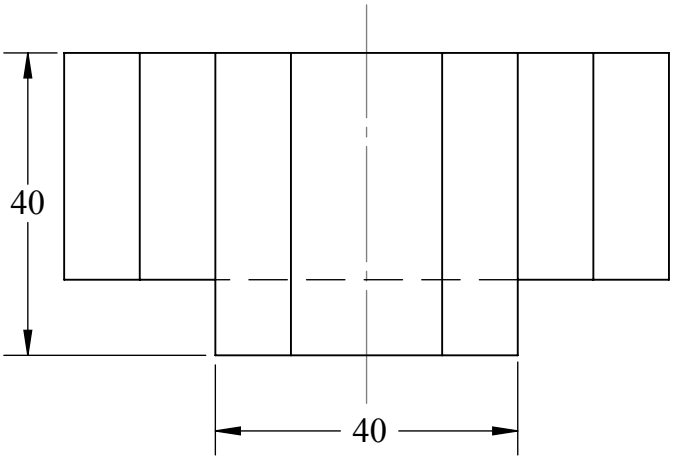
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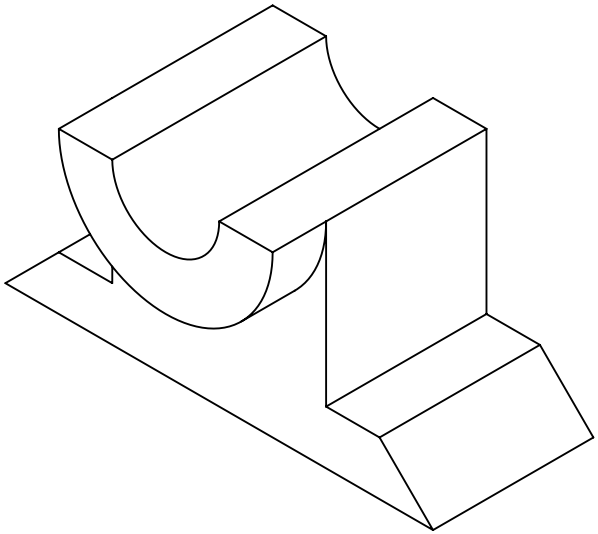
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

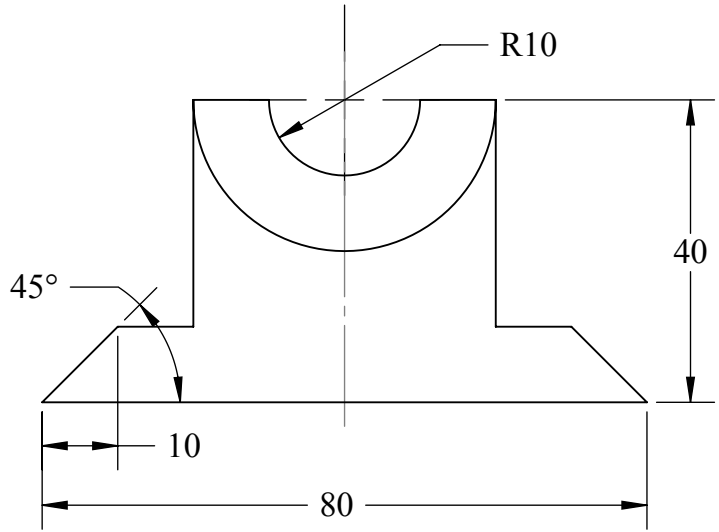
PART 01 | Default template material - leave Generic / default unchanged

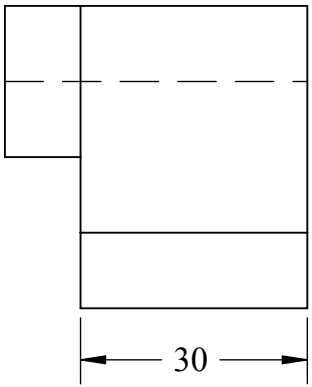
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 1

AUTODESK INVENTOR 2025 - LESSON 02

Rounded Base Bracket

SUGGESTED COMMANDS

Line | Arc | General Dimension | Extrude | Extrude Cut | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the primary side/base profile.
2. Extrude the profile to establish the overall width.
3. Create the stepped/ramped upper portion using a second sketch and Extrude.
4. Use the rounded end geometry from the top view to shape the front.
5. Use Hole for the 20 mm diameter through-hole.
6. Verify the R10 end and overall 70 x 40 relationships.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Multiple features, rounded geometry, Hole command.

MODEL REQUIREMENTS

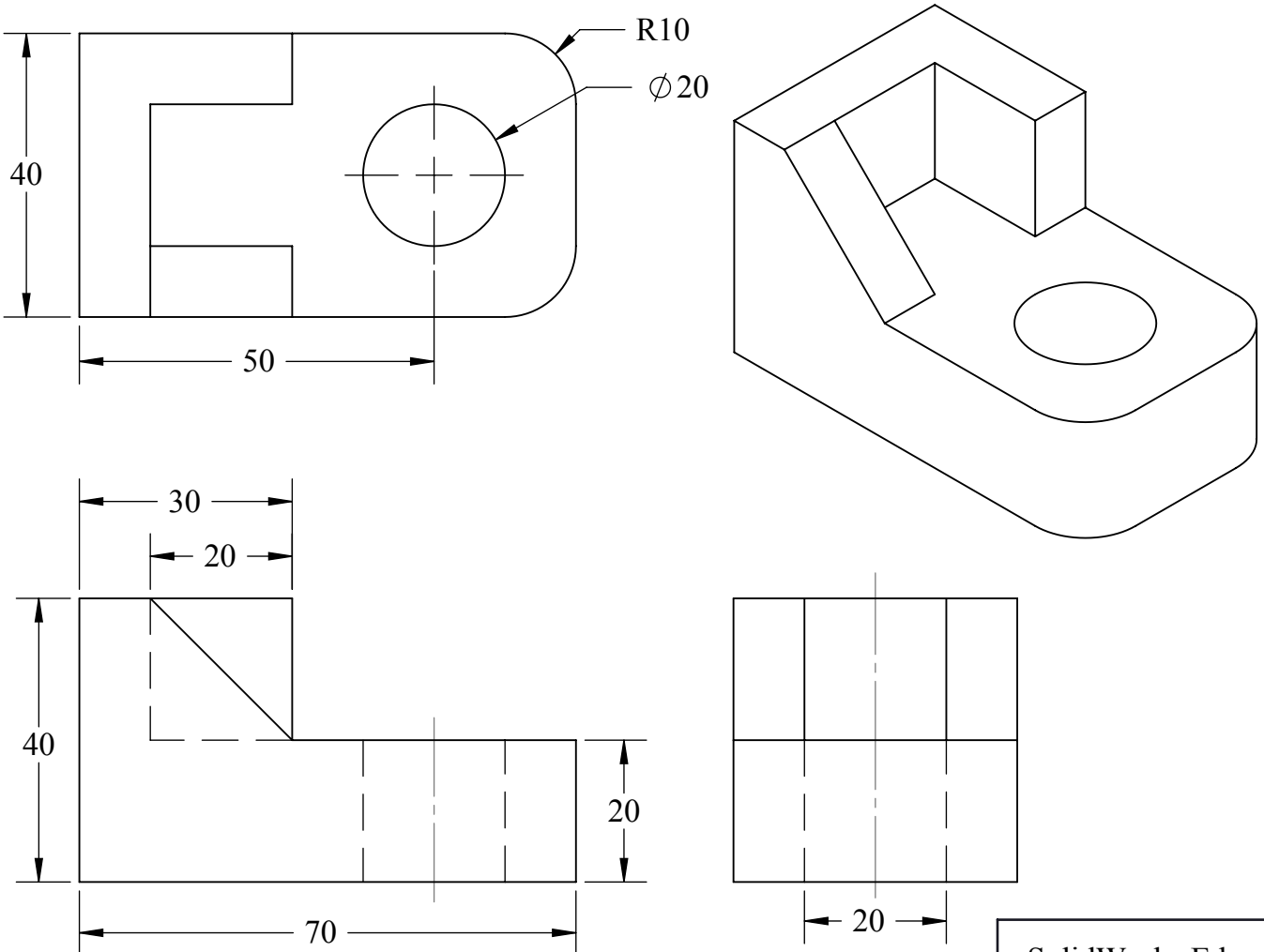
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_02.ipt | Verify Front, Top, Right, Isometric

ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

PART 02 | Default template material - leave Generic / default unchanged

- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.



SolidWorks Education
Lesson 2

AUTODESK INVENTOR 2025 - LESSON 03

Inclined Step Block

SUGGESTED COMMANDS

Line | General Dimension | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Select the view containing the inclined edge as the primary sketch.
2. Draw the complete closed profile with Line.
3. Dimension the 40, 20, and 10 mm geometry shown.
4. Fully constrain the sketch.
5. Extrude the profile to the required width.
6. Compare the resulting step and inclined face with the isometric view.

SUGGESTED COMPLETION TIME

Target time: 8 min Advanced challenge: ~6 min Extended time (2x): 16 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Simple profile extrusion and inclined lines.

MODEL REQUIREMENTS

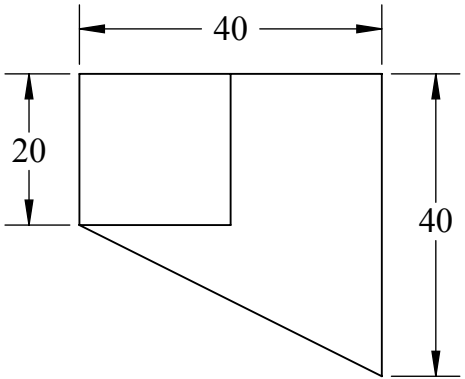
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_03.ipt | Verify Front, Top, Right, Isometric

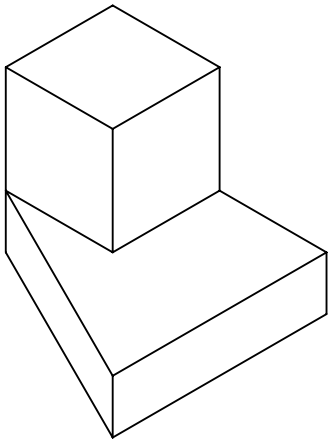
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

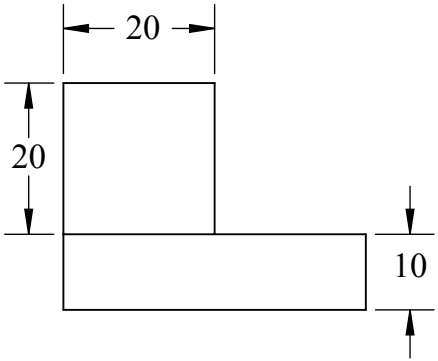
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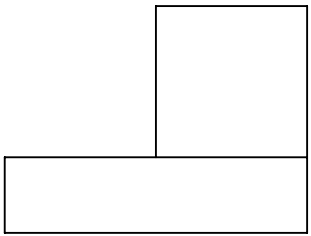
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 3

AUTODESK INVENTOR 2025 - LESSON 04

Rounded Lug Base

SUGGESTED COMMANDS

Rectangle | Circle | Arc | Trim | Extrude | Hole | Fillet

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the rectangular base.
2. Extrude the base to its specified thickness.
3. Sketch the raised rounded lug on the top/base face.
4. Use circles/arcs and Trim to form the rounded profile.
5. Extrude the raised portion.
6. Create the 20 mm diameter opening with Hole.
7. Apply R10 rounding where shown.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Boss creation, circles, trimming, fillets.

MODEL REQUIREMENTS

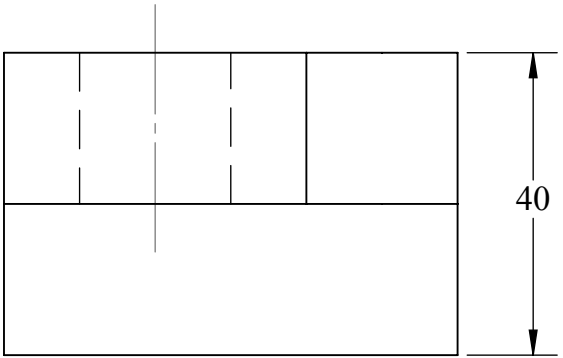
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_04.ipt | Verify Front, Top, Right, Isometric

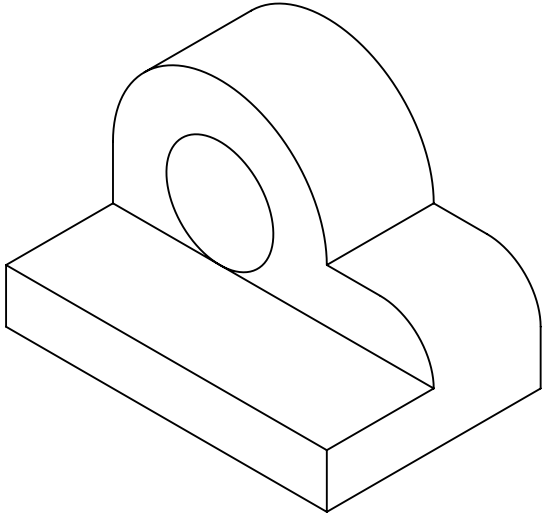
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

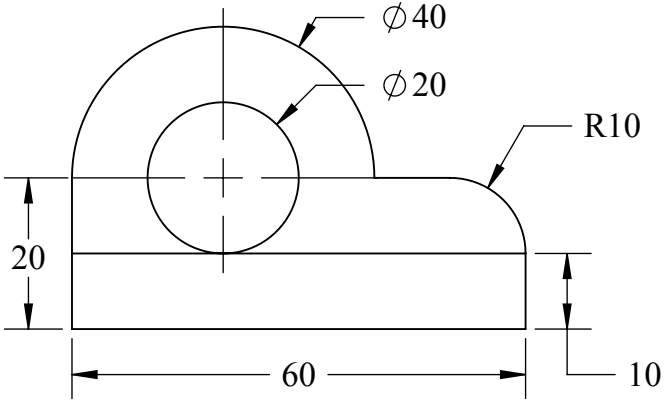
PART 04 | Default template material - leave Generic / default unchanged

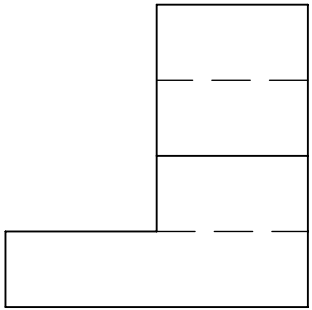
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 4

AUTODESK INVENTOR 2025 - LESSON 05

Frame with Curved Corners

SUGGESTED COMMANDS

Line | Arc | Sketch Fillet | Extrude | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the 10 mm base first.
2. Sketch the plan/profile of the vertical members.
3. Use Sketch Fillet or tangent arcs for the three R10 locations.
4. Extrude the vertical portions to the required heights.
5. Create the inclined support with Rib or a triangular Extrude.
6. Verify the 60 mm overall length and 50 mm height.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Fillets, multiple extrusions, structural ribs.

MODEL REQUIREMENTS

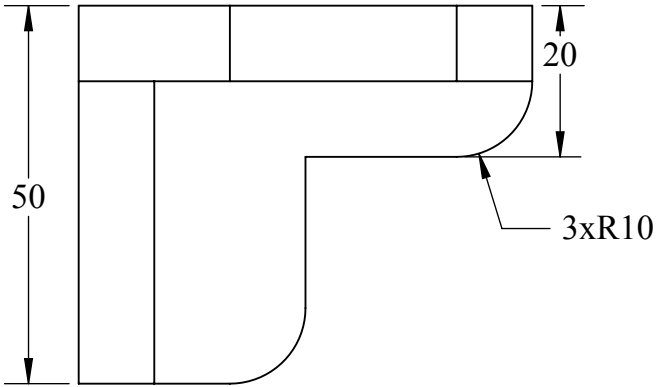
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_05.ipt | Verify Front, Top, Right, Isometric

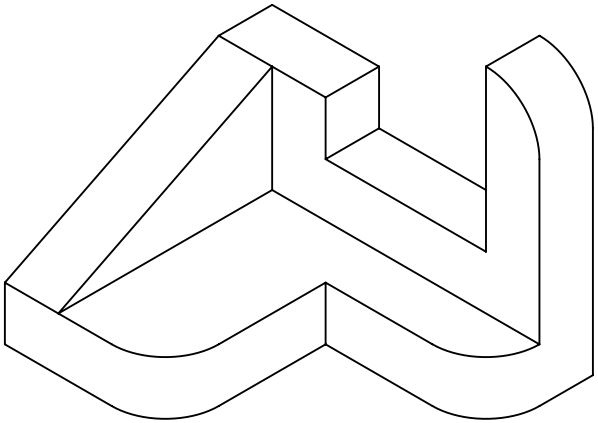
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

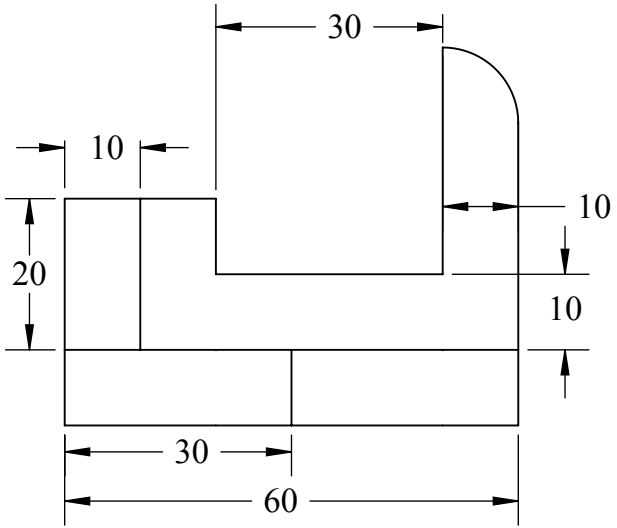
PART 05 | Default template material - leave Generic / default unchanged

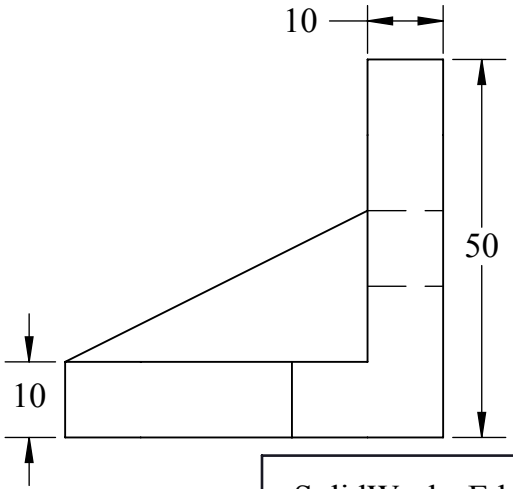
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 5

AUTODESK INVENTOR 2025 - LESSON 06

Stepped Rounded Block

SUGGESTED COMMANDS

Line | Arc | General Dimension | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the stepped side profile.
2. Enter the 60 mm overall length and 40 mm height.
3. Add each 10 mm step.
4. Include the rounded 40 mm diameter end using an arc.
5. Fully constrain the profile.
6. Extrude it to the depth shown in the orthographic views.

SUGGESTED COMPLETION TIME

Target time: 10 min Advanced challenge: ~8 min Extended time (2x): 20 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Combining stepped and circular geometry in one sketch.

MODEL REQUIREMENTS

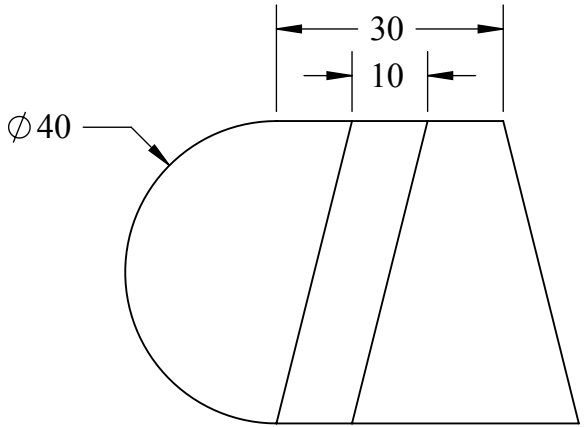
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_06.ipt | Verify Front, Top, Right, Isometric

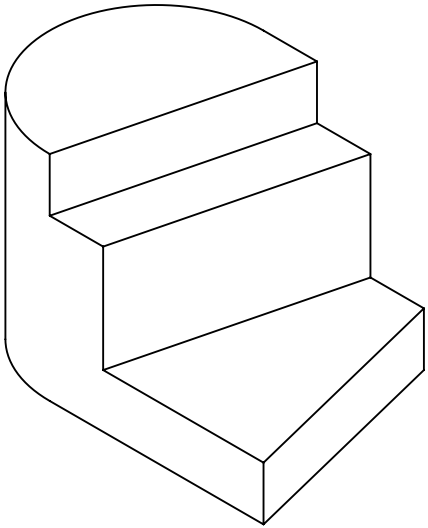
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

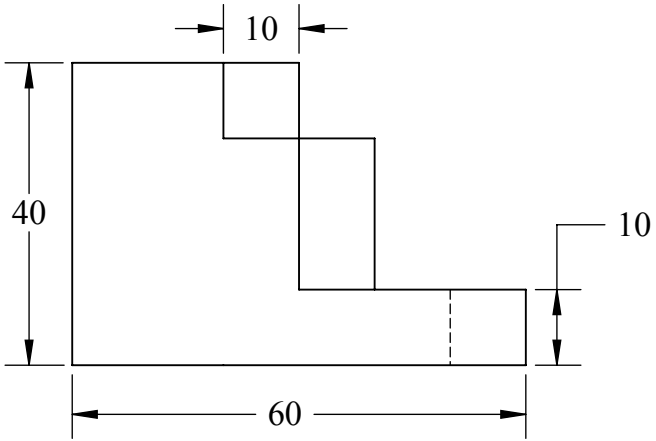
PART 06 | Default template material - leave Generic / default unchanged

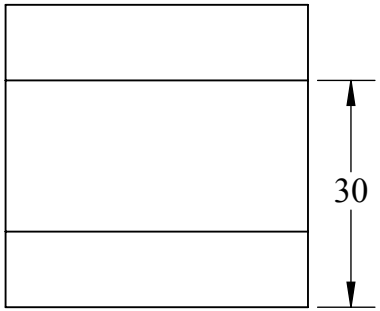
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 6

AUTODESK INVENTOR 2025 - LESSON 07

Base with Circular Support

SUGGESTED COMMANDS

Rectangle | Arc | Circle | Extrude | Hole | Fillet

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Extrude the rectangular base.
2. Sketch the rear upright profile.
3. Construct the 40 mm diameter outer circular feature and R10 transition.
4. Extrude the upright.
5. Create the concentric 20 mm diameter hole with Hole.
6. Add the smaller front support feature.
7. Verify the 60 mm base length and 10 mm feature thicknesses.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Concentric geometry, holes, tangent transitions.

MODEL REQUIREMENTS

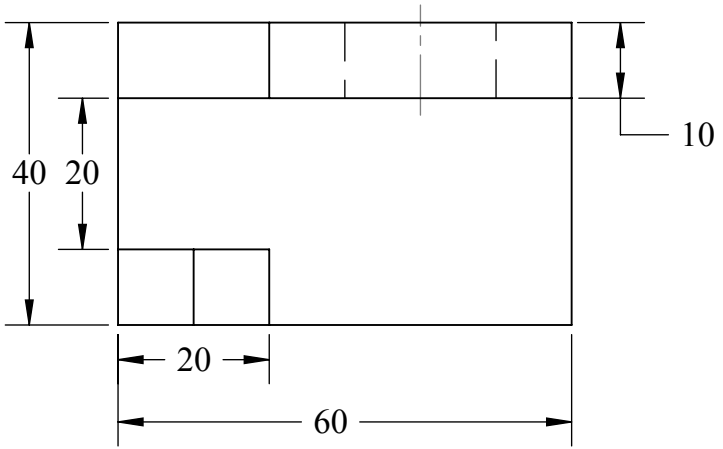
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_07.ipt | Verify Front, Top, Right, Isometric

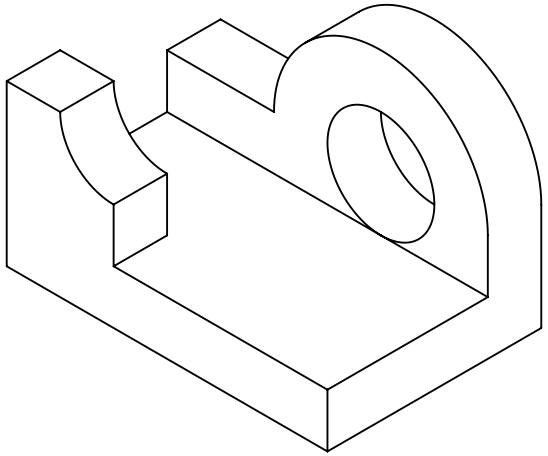
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

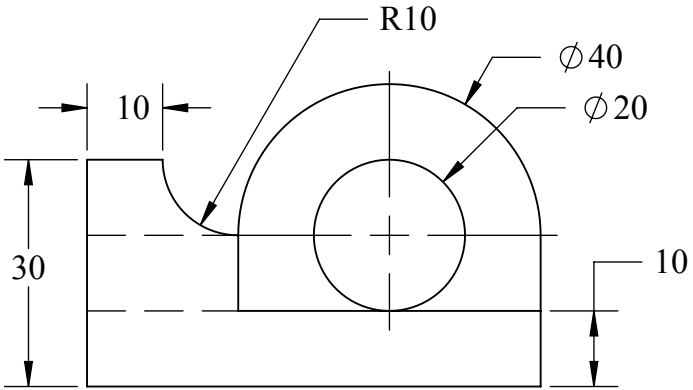
PART 07 | Default template material - leave Generic / default unchanged

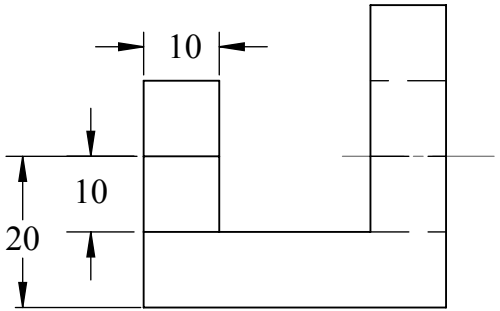
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 7

AUTODESK INVENTOR 2025 - LESSON 08

Arched Support

SUGGESTED COMMANDS

Line | Circle | Trim | Extrude | Extrude Cut | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the bottom base.
2. Build the rear circular/arched support using concentric circles.
3. Trim the circles to create the opening/notch shown.
4. Extrude the arch.
5. Create the base hole with Hole.
6. Add or remove the small upper block to match the drawing.
7. Inspect hidden edges against the orthographic views.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Concentric arcs, trims, cut features.

MODEL REQUIREMENTS

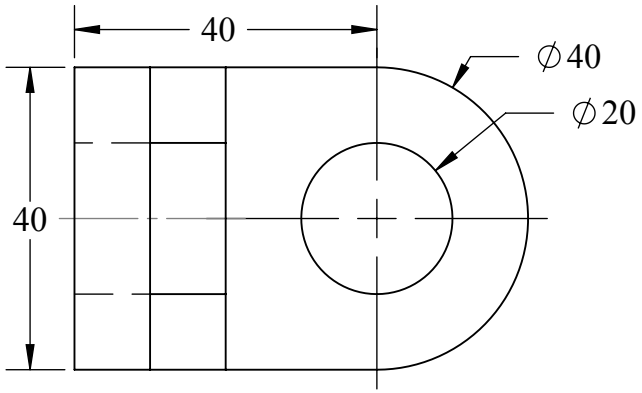
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_08.ipt | Verify Front, Top, Right, Isometric

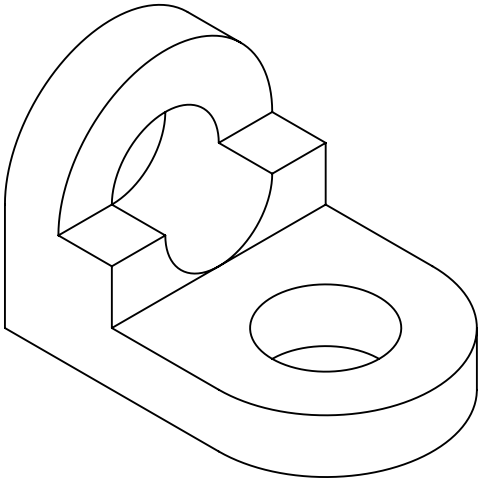
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

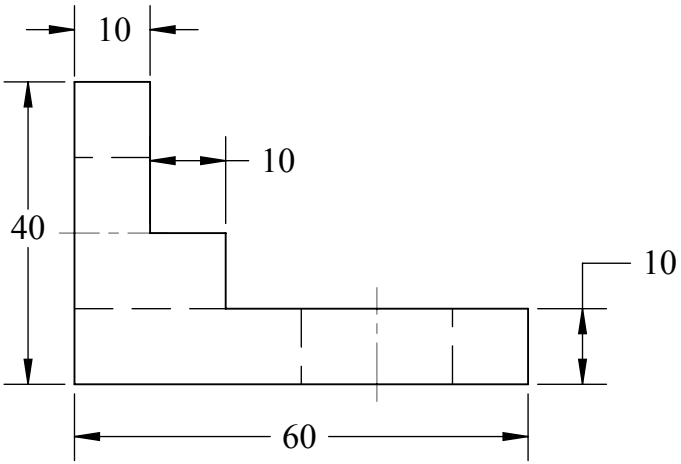
PART 08 | Default template material - leave Generic / default unchanged

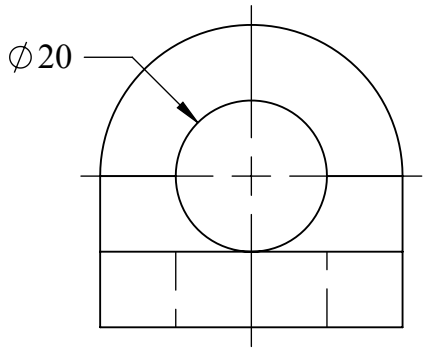
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 8

AUTODESK INVENTOR 2025 - LESSON 09

Twin Inclined Supports

SUGGESTED COMMANDS

Rectangle | Line | Extrude | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Extrude the rectangular base.
2. Sketch one inclined support profile.
3. Dimension the support using the 40/30/10 mm relationships shown.
4. Use Extrude or Rib to create the first inclined member.
5. Create the second member at the required location.
6. Check spacing and symmetry against the top/front views.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Sloping geometry and rib-like features.

MODEL REQUIREMENTS

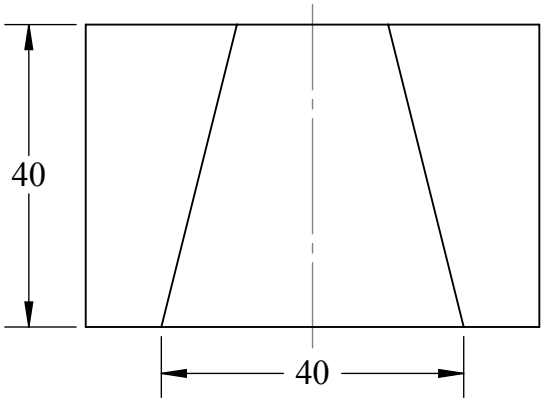
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_09.ipt | Verify Front, Top, Right, Isometric

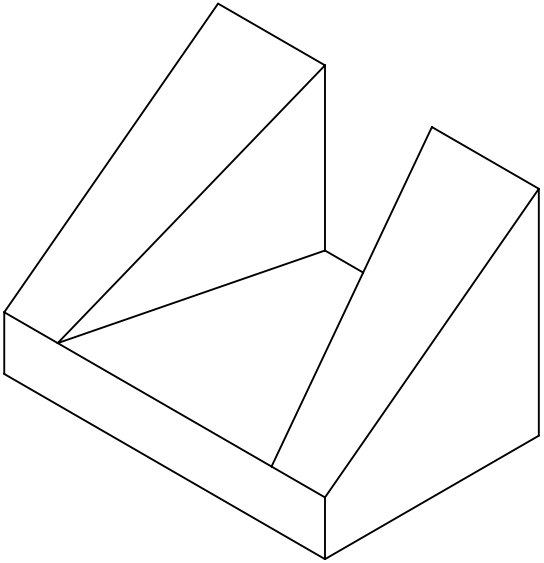
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

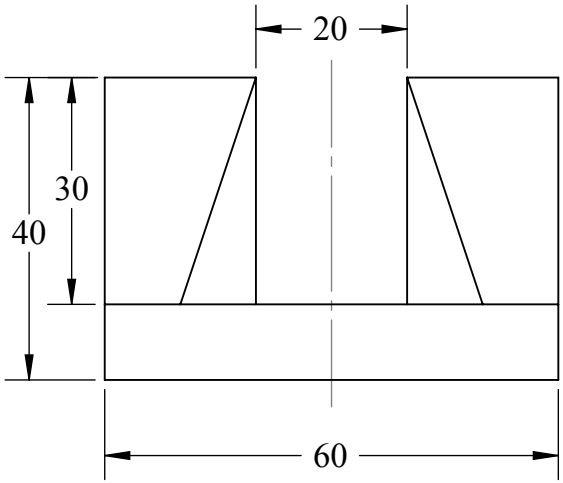
PART 09 | Default template material - leave Generic / default unchanged

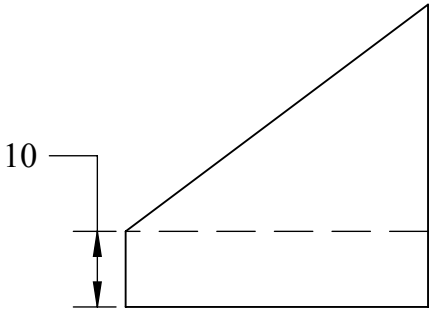
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 9

AUTODESK INVENTOR 2025 - LESSON 10

Angled Stepped Block

SUGGESTED COMMANDS

Line | General Dimension | Extrude | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the primary stepped profile.
2. Dimension the 40, 30, 20, and 10 mm steps.
3. Establish the 45 degree inclined geometry.
4. Extrude the completed profile.
5. Where practical, create the angled edge with Chamfer instead of sketching it.
6. Compare the front, top, and right-side views.

SUGGESTED COMPLETION TIME

Target time: 10 min Advanced challenge: ~8 min Extended time (2x): 20 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

45 degree geometry and chamfers.

MODEL REQUIREMENTS

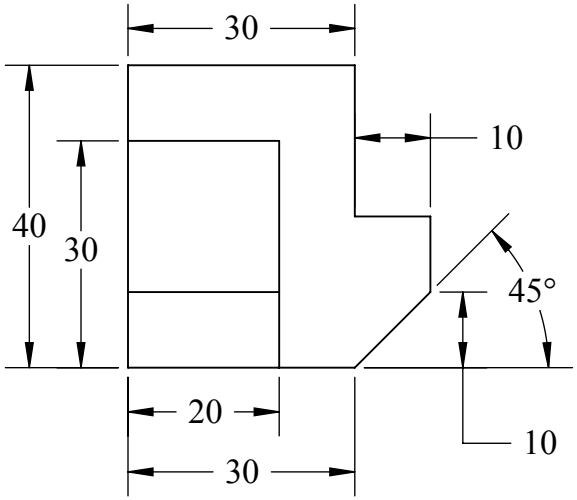
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_10.ipt | Verify Front, Top, Right, Isometric

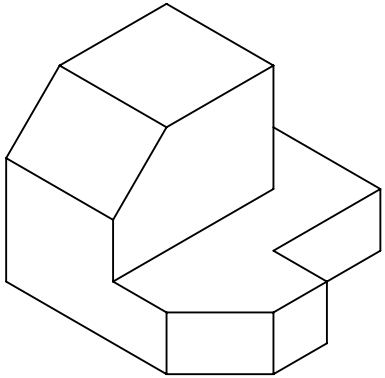
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

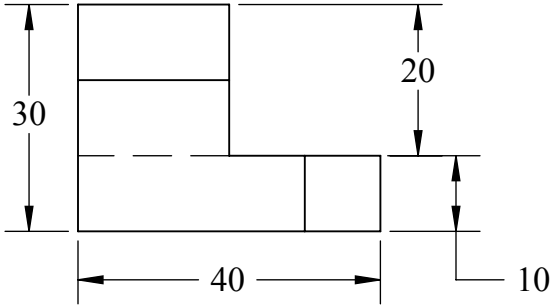
PART 10 | Default template material - leave Generic / default unchanged

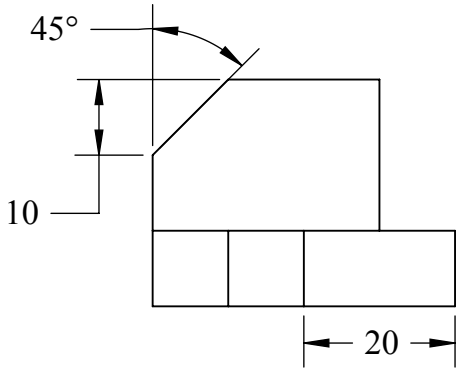
- No material assignment or iProperties grading for Lessons 01-10.
- Complete the original drawing geometry and verify dimensions and views.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 10

AUTODESK INVENTOR 2025 - LESSON 11

Multi-Level Block

SUGGESTED COMMANDS

Rectangle | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the main 60 x 40 footprint/body.
2. Add the tall rear block.
3. Add the lower front rail.
4. Create the intermediate 10/20/30 mm stepped features.
5. Use Extrude Cut where removing material is easier than adding it.
6. Verify the model from all three orthographic directions.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Feature order and additive/subtractive modeling.

MODEL REQUIREMENTS

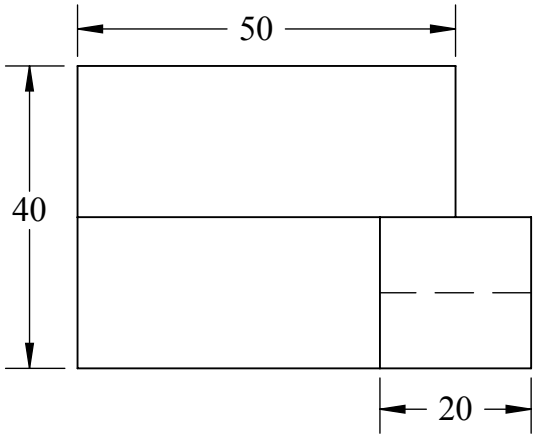
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_11.ipt | Verify Front, Top, Right, Isometric

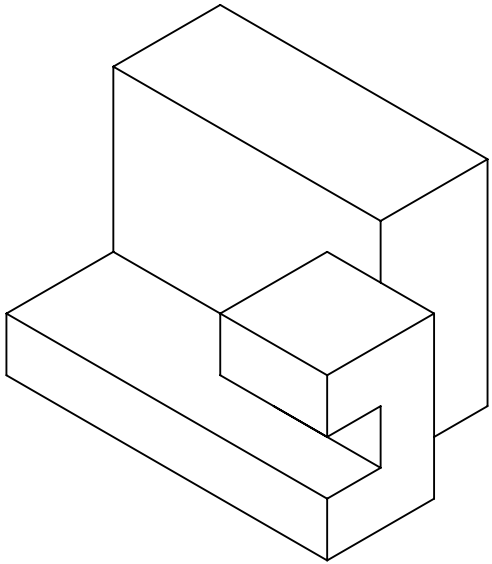
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

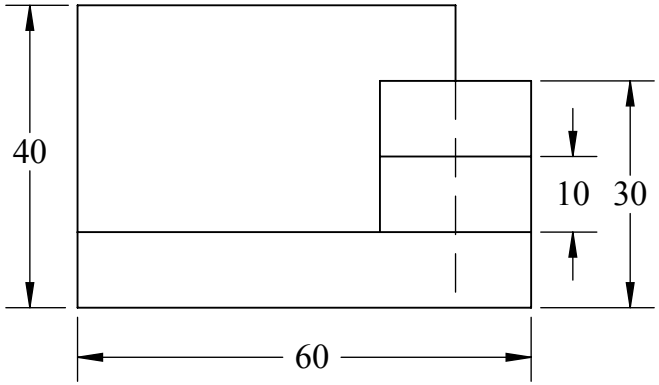
PART 11 | Assigned material: Steel, Mild

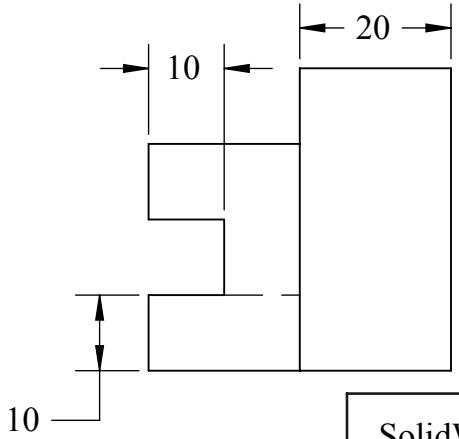
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 11

AUTODESK INVENTOR 2025 - LESSON 12

Angled Wall with Opening

SUGGESTED COMMANDS

Line | Extrude | Hole / Extrude Cut | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the 10 mm base.
2. Add the rear vertical wall.
3. Create the front/side raised wall.
4. Form the 45 degree angled face with a sketch or Chamfer.
5. Sketch the square opening on the upright wall.
6. Extrude Cut - Through All.
7. Check the 60 mm overall dimension.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Wall features, square cutouts, 45 degree geometry.

MODEL REQUIREMENTS

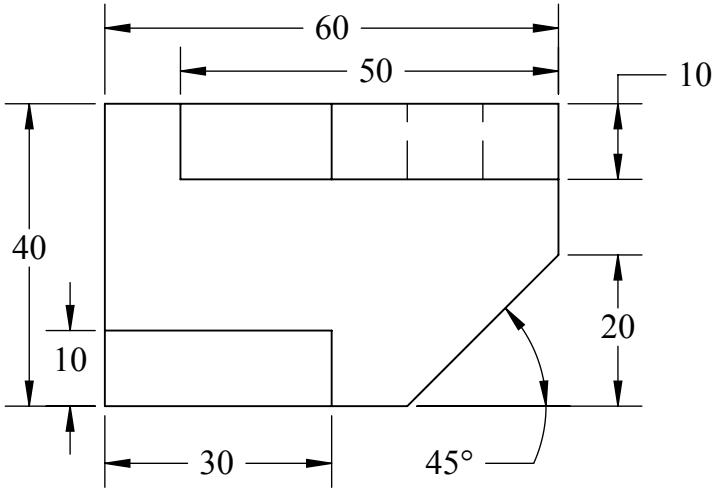
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_12.ipt | Verify Front, Top, Right, Isometric

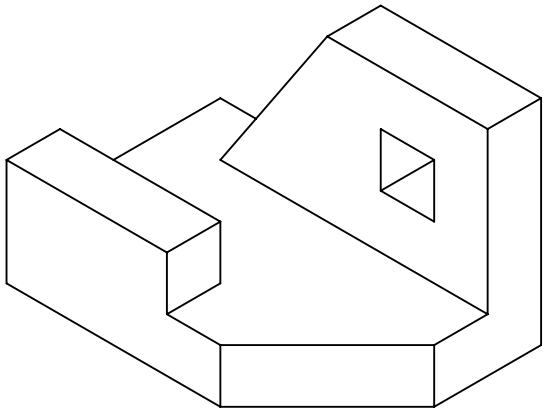
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

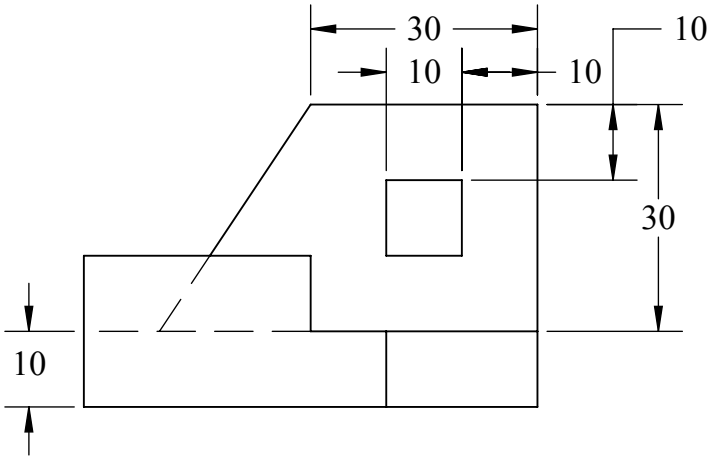
PART 12 | Assigned material: ABS Plastic

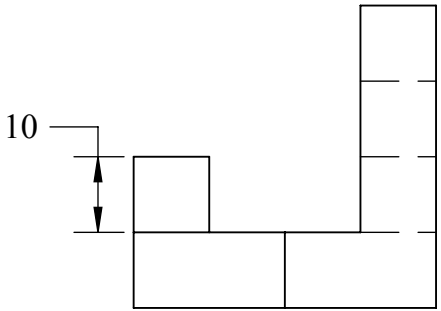
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 12

AUTODESK INVENTOR 2025 - LESSON 13

Hipped / Roof Form

SUGGESTED COMMANDS

Rectangle | Extrude | Work Plane | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the rectangular lower block.
2. Establish the roof starting height.
3. Sketch the primary roof slope from one orthographic direction.
4. Use Extrude Cut to remove the first wedge.
5. Sketch the second roof direction on a perpendicular face.
6. Make a second wedge cut.
7. Verify the 45 degree relationship and ridge position.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Building complex geometry through intersecting cuts.

MODEL REQUIREMENTS

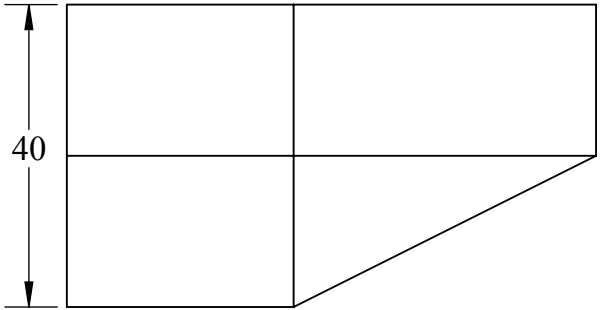
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_13.ipt | Verify Front, Top, Right, Isometric

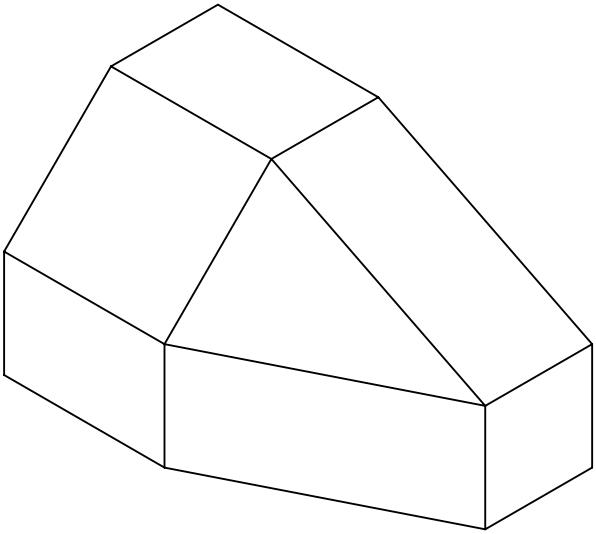
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

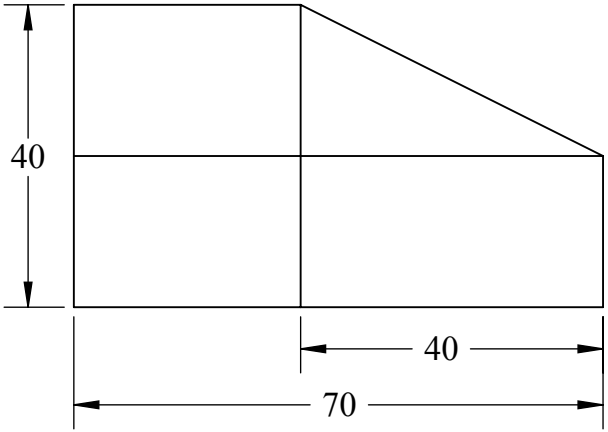
PART 13 | Assigned material: Stainless Steel

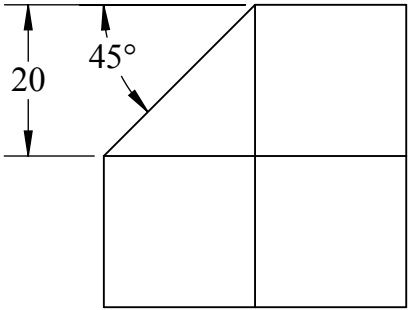
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 13

AUTODESK INVENTOR 2025 - LESSON 14

Multi-Slope Roof Block

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Make the full rectangular bounding block first.
2. Sketch one diagonal roof line on a side face.
3. Remove the upper triangular region.
4. Repeat from the perpendicular face.
5. Use the orthographic views to locate the ridge accurately.
6. Verify the final sloping faces in isometric view.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Triangular cut-extrudes.

MODEL REQUIREMENTS

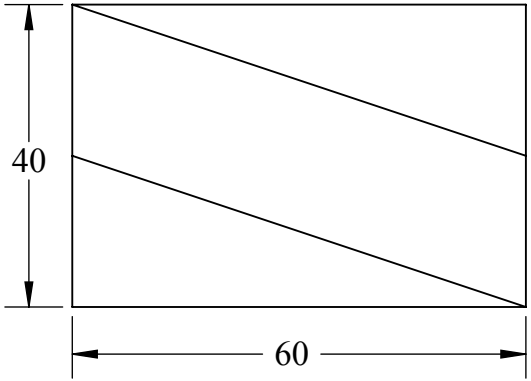
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_14.ipt | Verify Front, Top, Right, Isometric

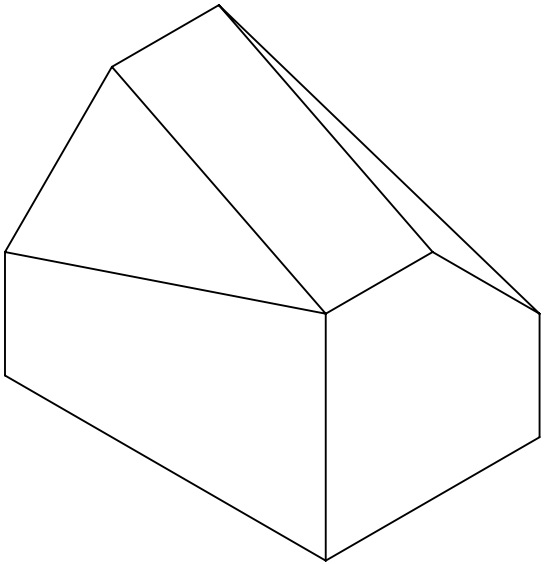
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

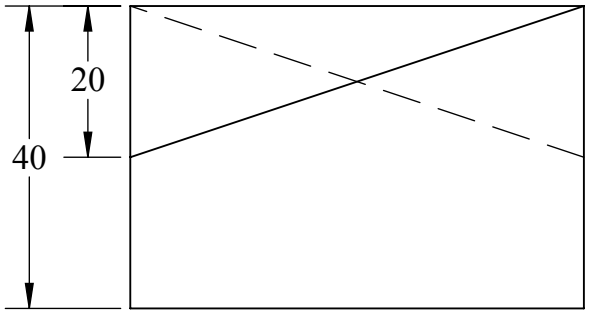
PART 14 | Assigned material: Acetal Resin, White

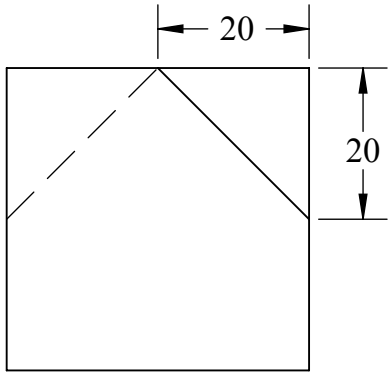
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 14

AUTODESK INVENTOR 2025 - LESSON 15

Stair-Step Pocket

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Extrude Cut | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the complete outer block/profile.
2. Establish the 45 degree outer inclined side.
3. Sketch the staircase pocket on the appropriate face.
4. Create the repeated 10 mm step geometry.
5. Extrude Cut the staircase through the required distance.
6. Inspect each level against the front/right views.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Stepped cuts and feature sequencing.

MODEL REQUIREMENTS

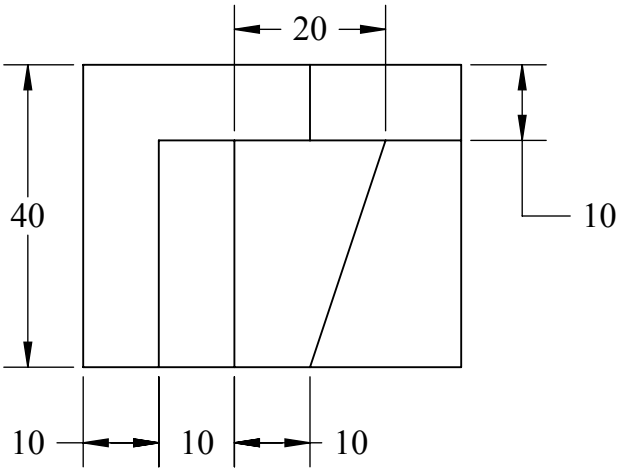
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_15.ipt | Verify Front, Top, Right, Isometric

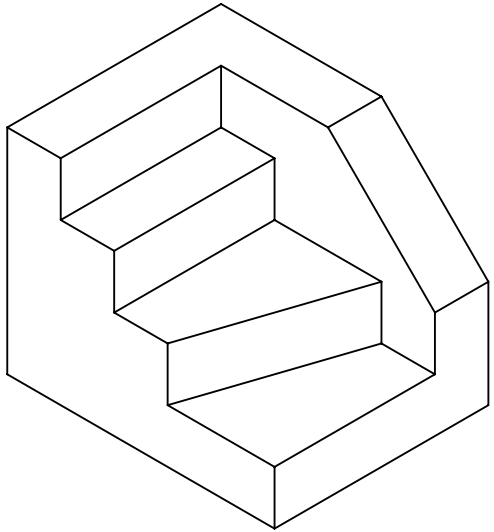
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

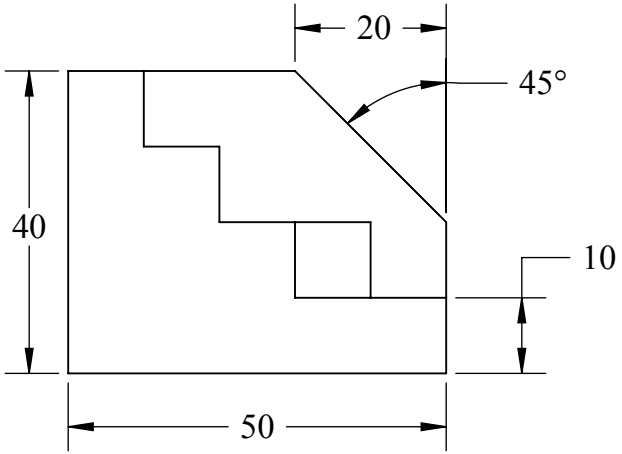
PART 15 | Assigned material: Aluminum 6061

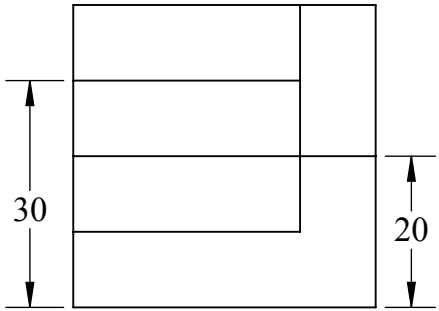
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 15

AUTODESK INVENTOR 2025 - LESSON 16

Faceted Wedge

SUGGESTED COMMANDS

Box / Rectangle | Extrude | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Begin with the maximum 60 x 40 bounding solid.
2. Use the front view to sketch the first diagonal cutting profile.
3. Cut the first triangular wedge.
4. Sketch a second diagonal profile from the perpendicular view.
5. Remove the second wedge.
6. Continue until the intersecting triangular faces match the isometric drawing.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Creating faceted solids from multiple cutting directions.

MODEL REQUIREMENTS

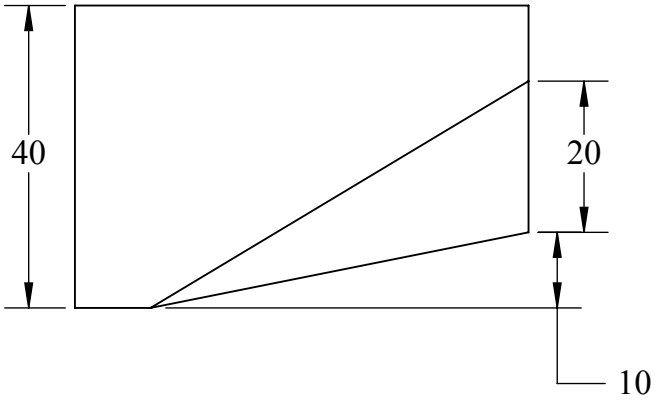
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_16.ipt | Verify Front, Top, Right, Isometric

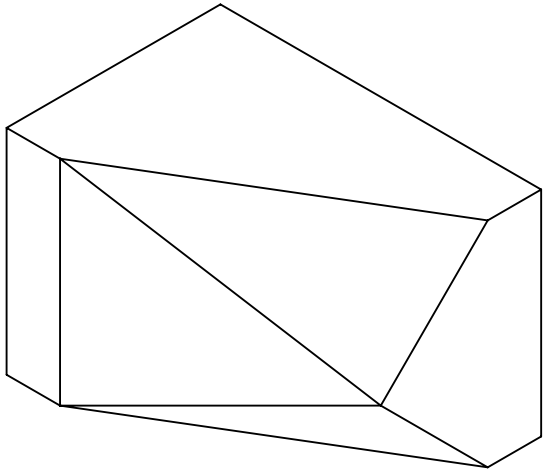
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

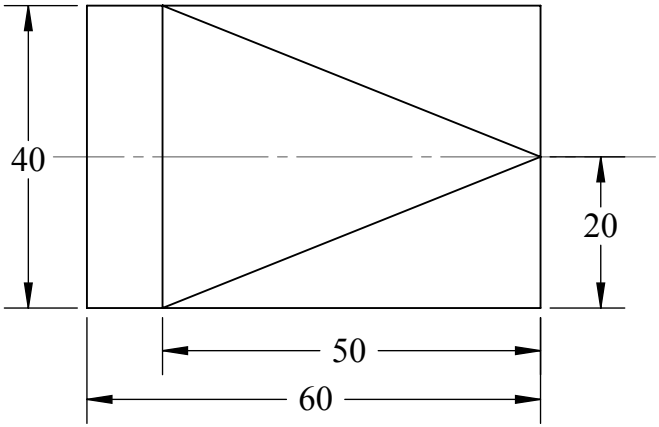
PART 16 | Assigned material: Steel, Mild

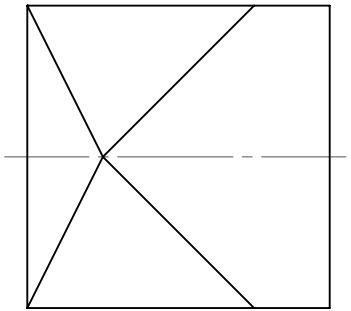
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 16

AUTODESK INVENTOR 2025 - LESSON 17

Interlocking H Block

SUGGESTED COMMANDS

Rectangle | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Start from a rectangular solid representing the maximum envelope.
2. Cut the upper rectangular slot.
3. Cut the lower slot/recess from the opposite face.
4. Create the central opening.
5. Use the 10 and 20 mm offsets to position each cut.
6. Check hidden-line locations carefully.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Orthographic visualization and subtractive modeling.

MODEL REQUIREMENTS

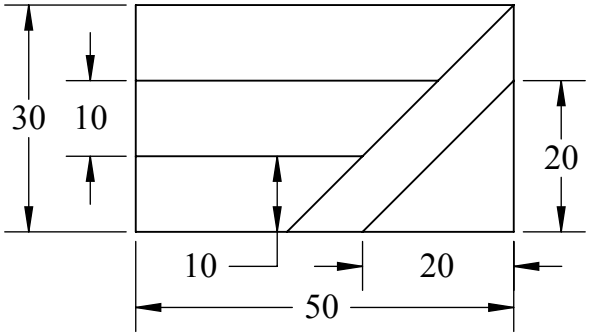
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_17.ipt | Verify Front, Top, Right, Isometric

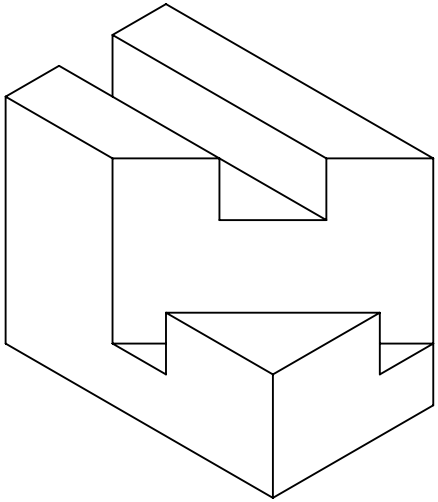
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

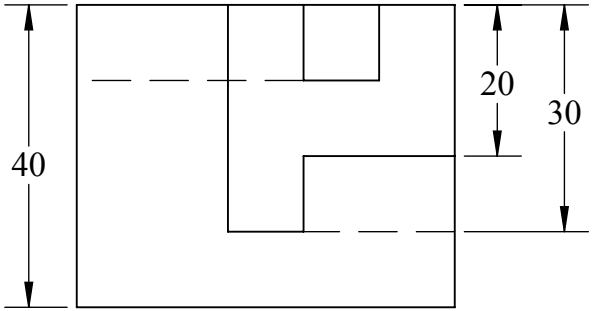
PART 17 | Assigned material: ABS Plastic

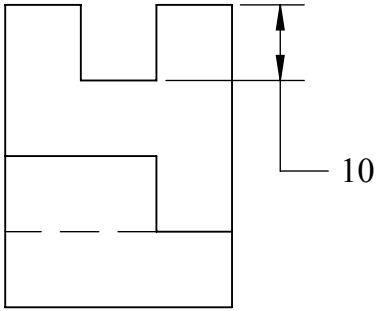
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 17

AUTODESK INVENTOR 2025 - LESSON 18

Curved C-Bracket

SUGGESTED COMMANDS

Line | Arc | Fillet | Extrude | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the C-shaped primary profile.
2. Create R20 outer geometry and the indicated R10 transitions.
3. Extrude the body.
4. Add the upper overhanging feature.
5. Create the 20 mm diameter hole.
6. Apply remaining R10 fillets.
7. Compare the upper and lower curved portions.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Tangent radii and curved bracket construction.

MODEL REQUIREMENTS

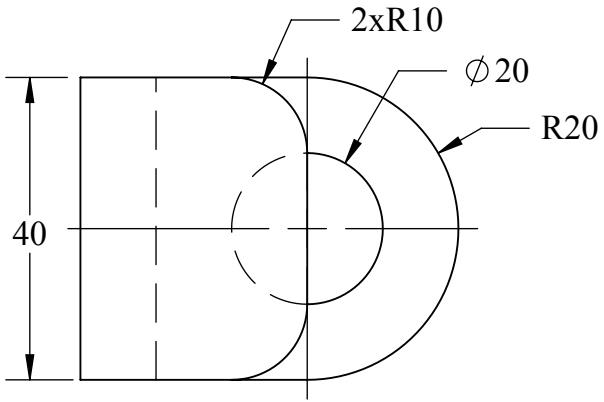
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_18.ipt | Verify Front, Top, Right, Isometric

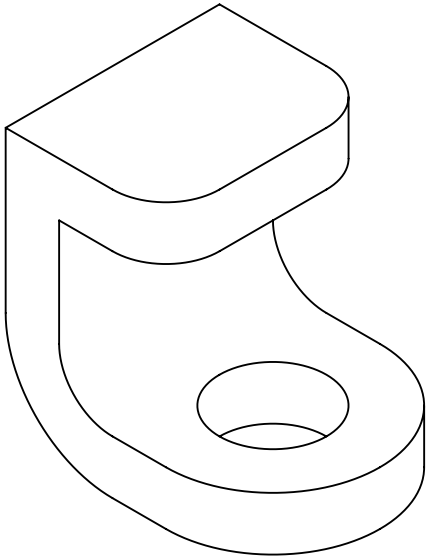
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

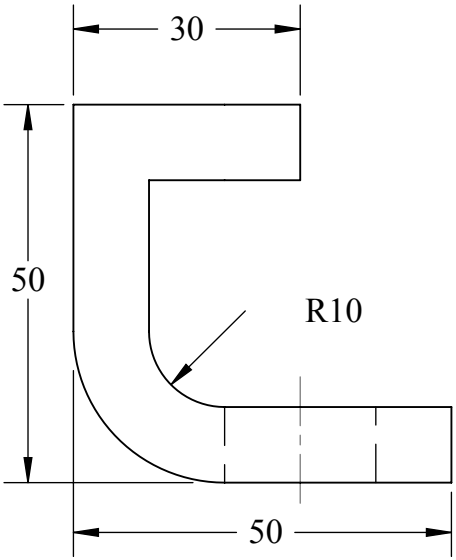
PART 18 | Assigned material: Acetal Resin, Black

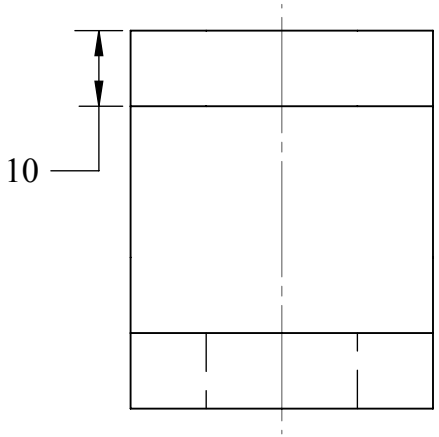
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 18

AUTODESK INVENTOR 2025 - LESSON 19

Internal Stair Frame

SUGGESTED COMMANDS

Line | Extrude | Extrude Cut | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the full outside block.
2. Sketch the internal stepped/open profile.
3. Cut the first pocket.
4. Add the 10 mm and 20 mm stepped levels.
5. Create the 45 degree inclined face.
6. Continue cutting until the internal geometry matches the isometric view.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Complex pockets and angled cuts.

MODEL REQUIREMENTS

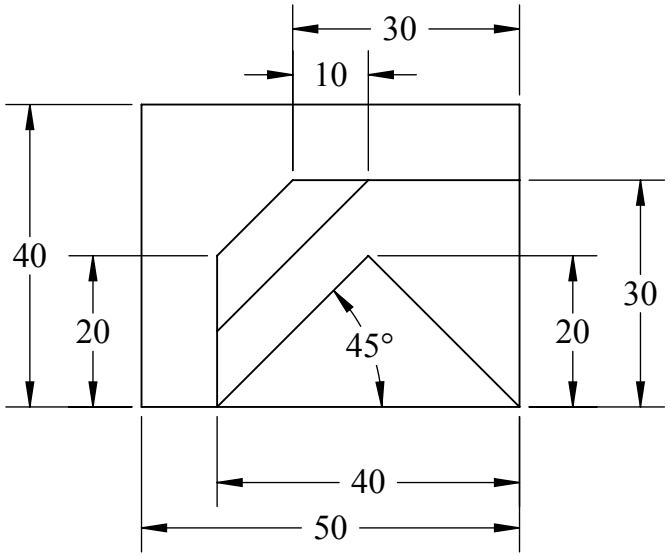
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_19.ipt | Verify Front, Top, Right, Isometric

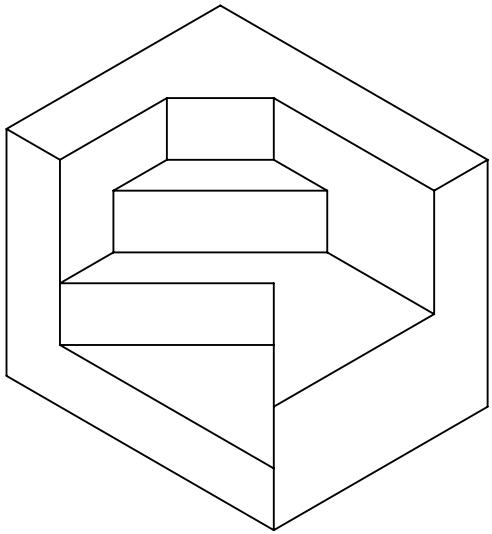
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

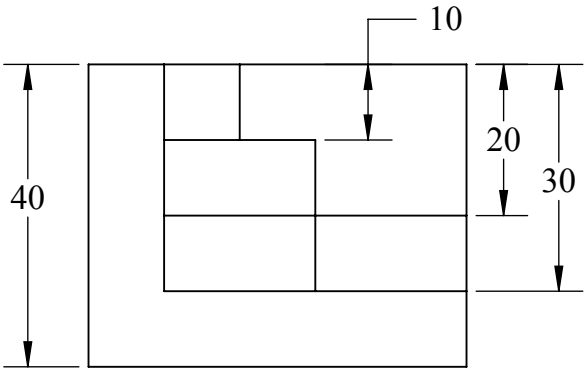
PART 19 | Assigned material: Acetal Resin, White

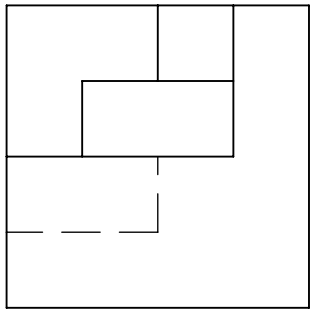
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 19

AUTODESK INVENTOR 2025 - LESSON 20

Rounded Base with Raised Supports

SUGGESTED COMMANDS

Arc | Circle | Extrude | Hole | Chamfer | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the rounded base profile using the R20 end.
2. Extrude the base.
3. Create the 20 mm diameter center hole.
4. Add the two vertical supports.
5. Create the central inclined/peaked feature with Rib or an Extrude.
6. Apply the 45 degree geometry.
7. Verify all 10 mm offsets.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Combined circular, hole, rib, and angled features.

MODEL REQUIREMENTS

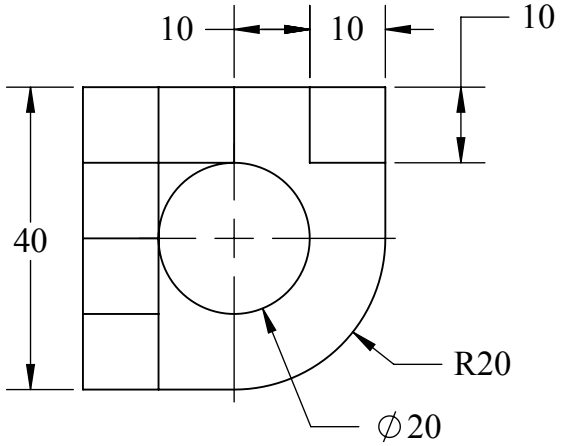
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_20.ipt | Verify Front, Top, Right, Isometric

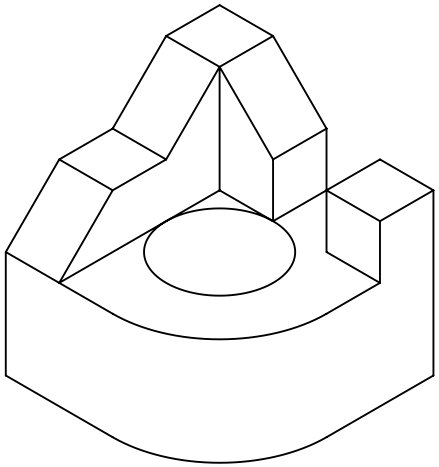
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

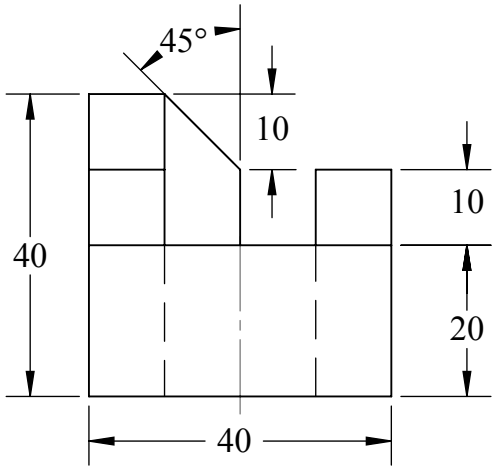
PART 20 | Assigned material: Steel, Mild

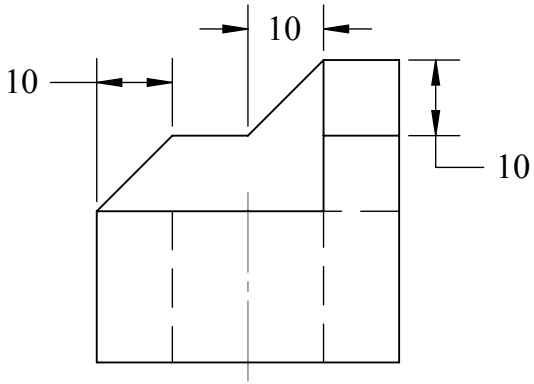
- Assign the specified physical Material (not just Appearance) from the Inventor Material Library.
- Save the IPT with the selected material; no iProperties submission required yet.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 20

AUTODESK INVENTOR 2025 - LESSON 21

Contoured Support

SUGGESTED COMMANDS

Line | Arc | Sketch Fillet | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the primary 70 mm profile.
2. Form the R20 rounded end.
3. Extrude the full body.
4. Use side-profile cuts to establish the stepped areas.
5. Create the two R10 internal transitions.
6. Confirm the 50 mm height and 40 mm width relationships.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Compound radii and profile cuts.

MODEL REQUIREMENTS

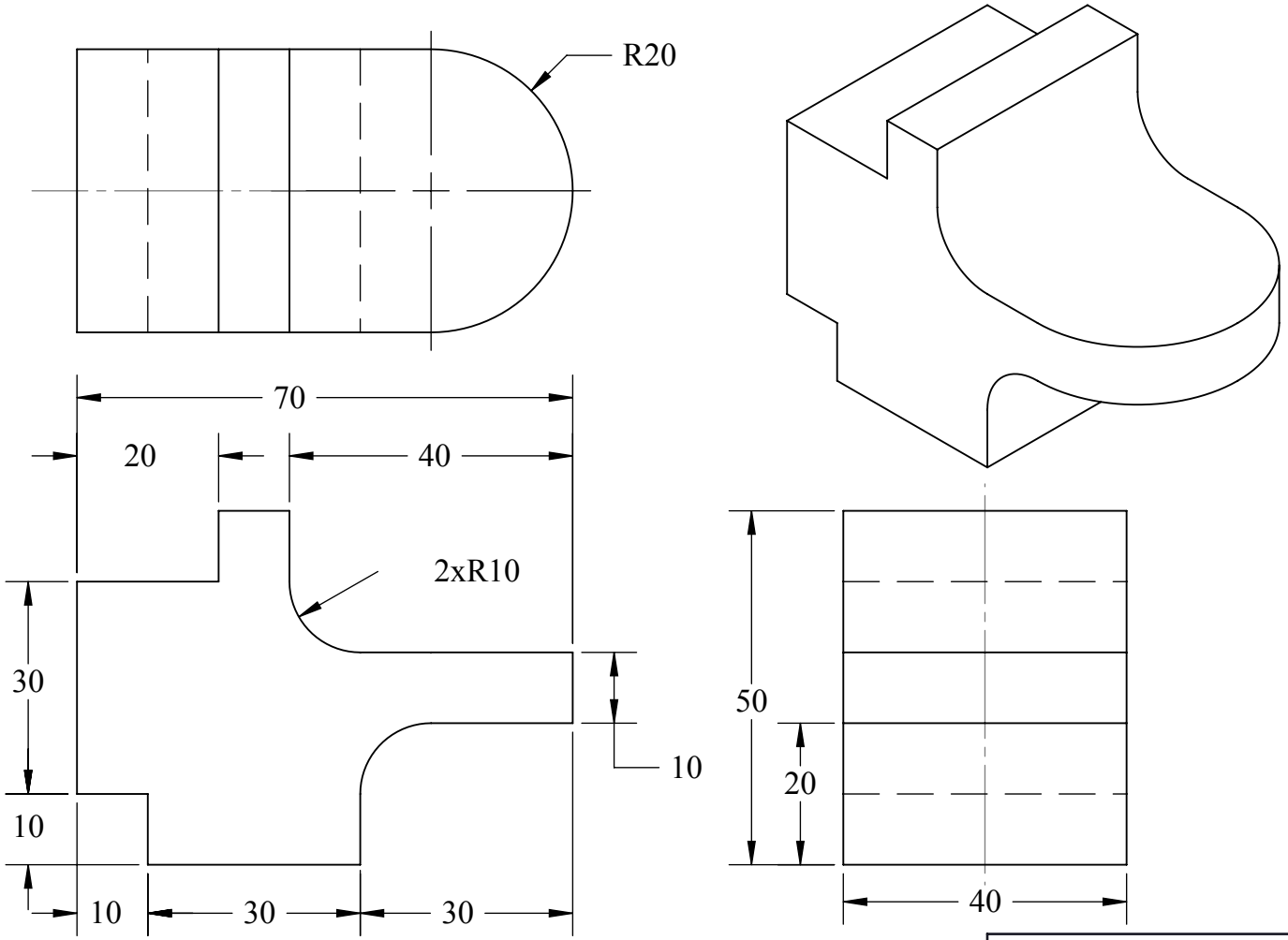
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_21.ipt | Verify Front, Top, Right, Isometric

ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

PART 21 | Assigned material: Aluminum 6061

- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.



SolidWorks Education
Lesson 21

AUTODESK INVENTOR 2025 - LESSON 22

Two-Hole Curved Bracket

SUGGESTED COMMANDS

Arc | Extrude | Fillet | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the vertical bracket/body.
2. Add the projecting horizontal arm.
3. Round the front arm using R20.
4. Create the first 20 mm diameter hole in the vertical portion.
5. Create the second 20 mm diameter hole in the horizontal portion.
6. Apply remaining transitions/fillets.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Holes on perpendicular faces.

MODEL REQUIREMENTS

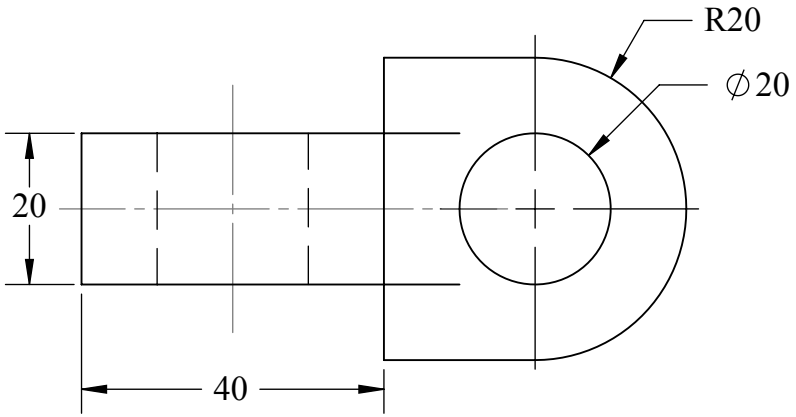
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_22.ipt | Verify Front, Top, Right, Isometric

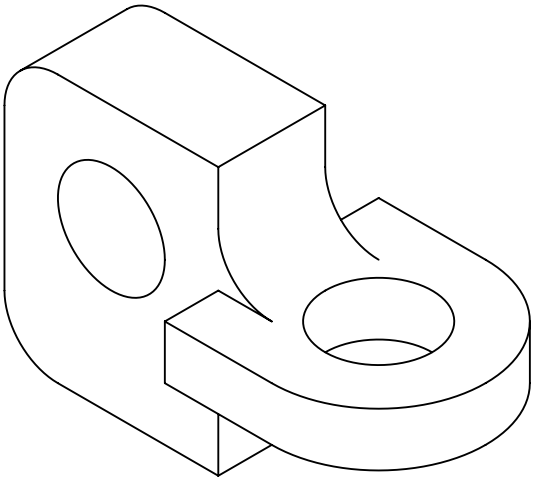
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

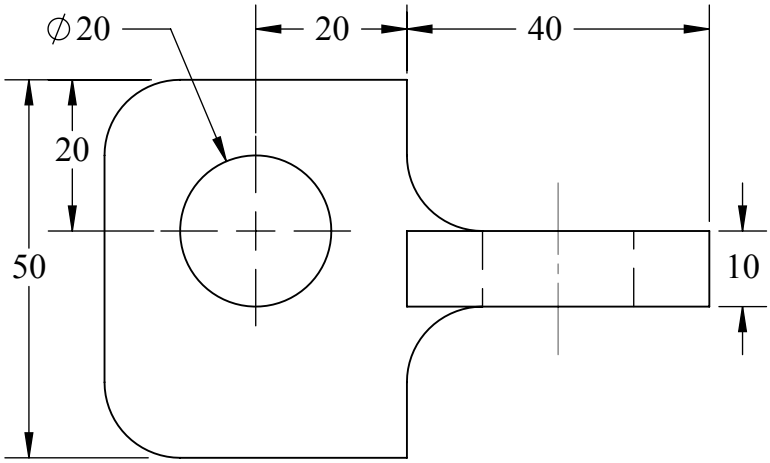
PART 22 | Assigned material: ABS Plastic

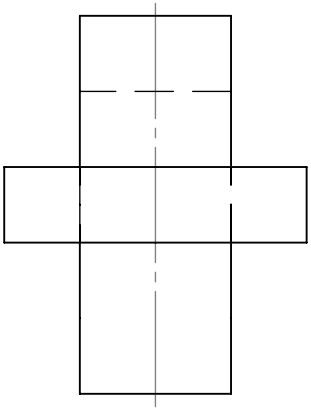
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 22

AUTODESK INVENTOR 2025 - LESSON 23

Internal Staircase

SUGGESTED COMMANDS

Rectangle | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Start with the maximum rectangular block.
2. Sketch the first 10 mm-deep step pocket.
3. Extrude Cut the first step.
4. Repeat for the second and third levels.
5. Use the 10, 20 and 30 mm offsets shown.
6. Check the stair progression in the isometric view.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Repeated stepped pockets.

MODEL REQUIREMENTS

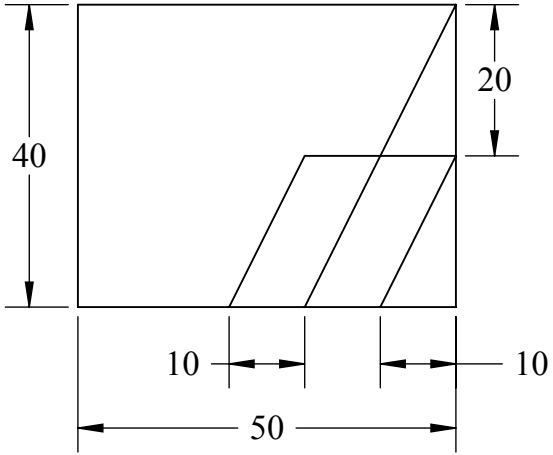
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_23.ipt | Verify Front, Top, Right, Isometric

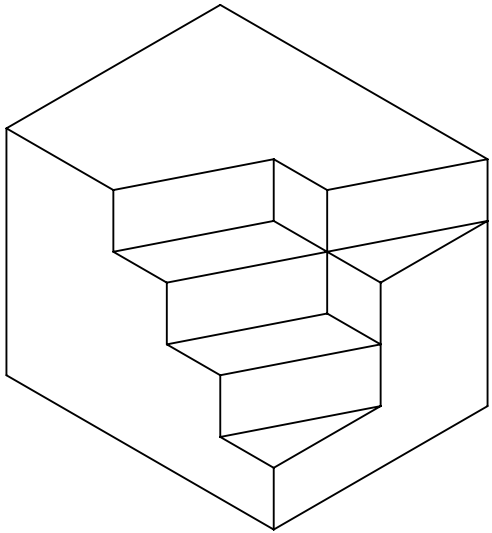
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

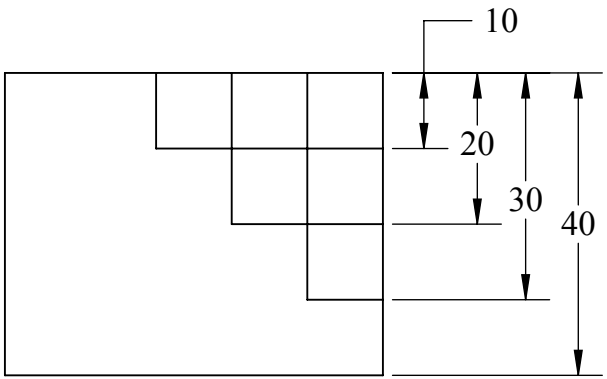
PART 23 | Assigned material: Acetal Resin, White

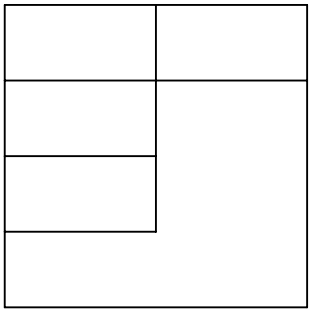
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 23

AUTODESK INVENTOR 2025 - LESSON 24

Inverted Pyramidal / V-Groove Form

SUGGESTED COMMANDS

Rectangle | Extrude | Work Plane | Loft Cut | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the full rectangular block.
2. Locate the convergence point using the orthographic views.
3. Create the top opening/profile.
4. Use Loft Cut from the opening toward the required point/edge.
5. If Loft is not being taught, create the faces with several triangular Extrude Cuts.
6. Verify all converging edges.

SUGGESTED COMPLETION TIME

Target time: 22 min Advanced challenge: ~16 min Extended time (2x): 44 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Loft Cut and complex planar surfaces.

MODEL REQUIREMENTS

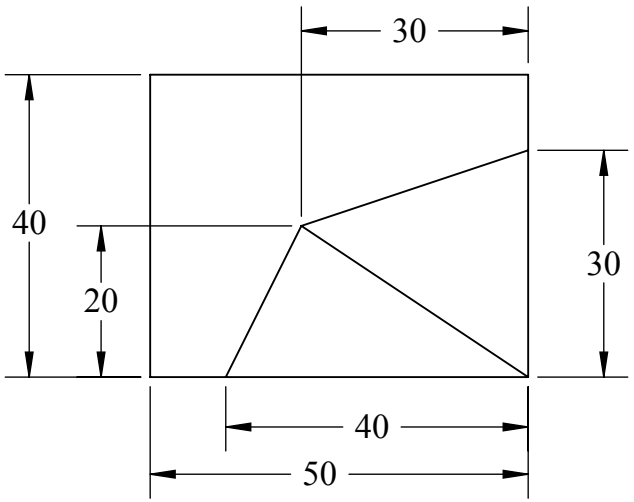
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_24.ipt | Verify Front, Top, Right, Isometric

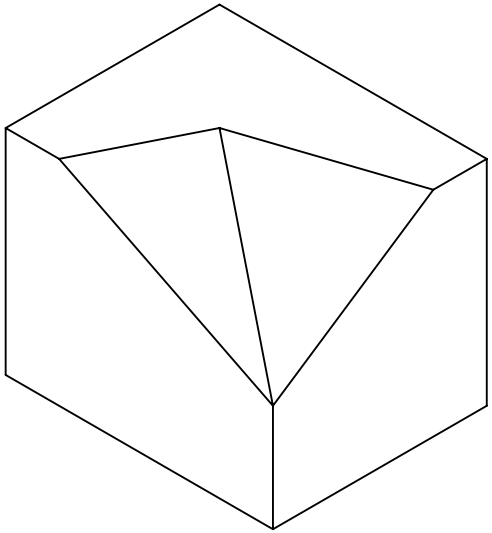
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

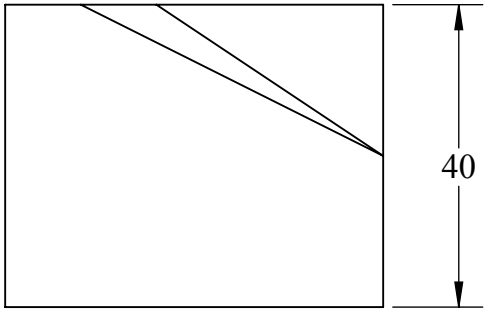
PART 24 | Assigned material: Acetal Resin, White

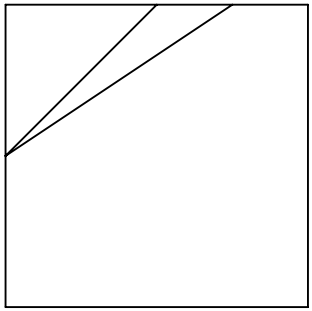
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 24

AUTODESK INVENTOR 2025 - LESSON 25

Two-Lug Angle Bracket

SUGGESTED COMMANDS

Arc | Extrude | Hole | Fillet

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the bottom mounting plate.
2. Form the rounded end.
3. Create its 20 mm diameter hole.
4. Sketch and extrude the tall vertical plate.
5. Include the large rounded 40 mm diameter outer end.
6. Create the second 20 mm diameter hole.
7. Confirm the 10 mm material thickness.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Perpendicular plates and aligned holes.

MODEL REQUIREMENTS

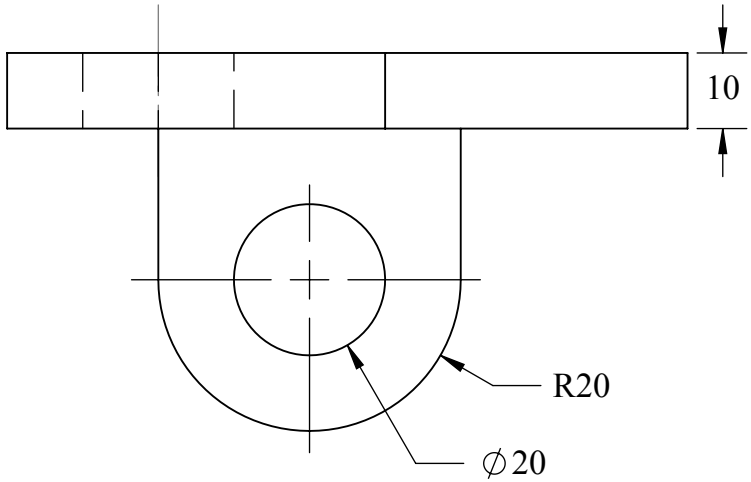
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_25.ipt | Verify Front, Top, Right, Isometric

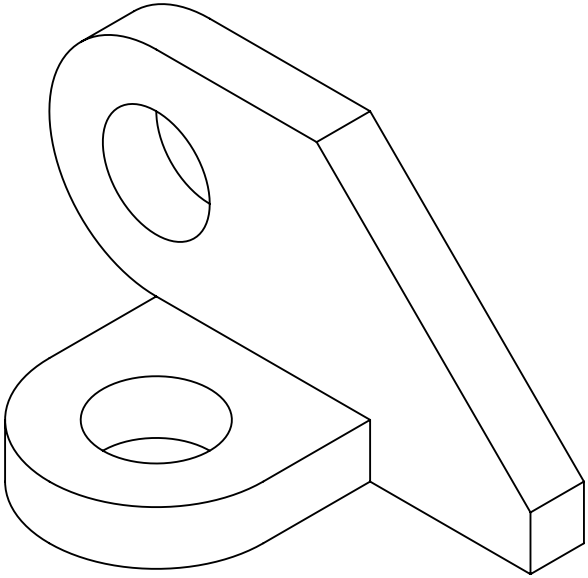
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

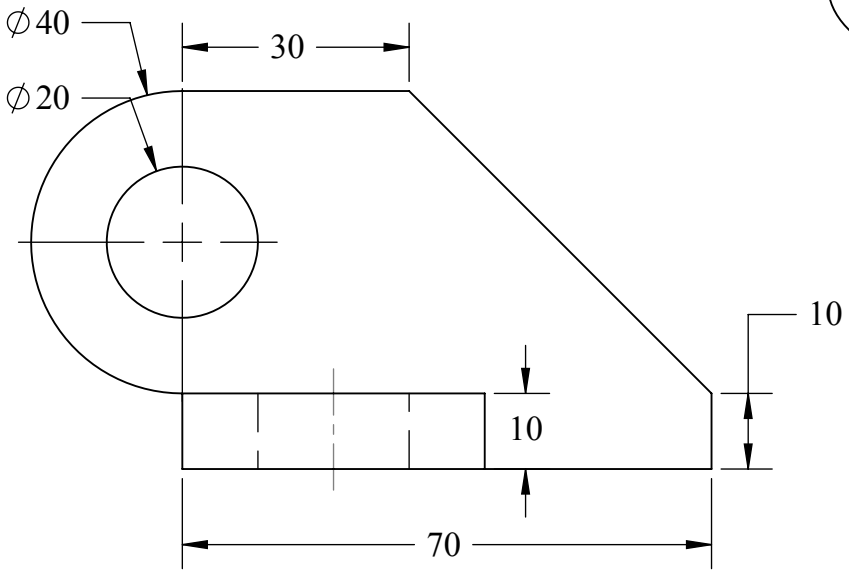
PART 25 | Assigned material: ABS Plastic

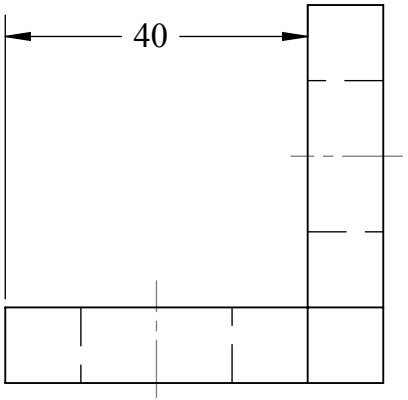
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 25

AUTODESK INVENTOR 2025 - LESSON 26

Base, Post and Gusset

SUGGESTED COMMANDS

Extrude | Hole | Fillet | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the 50 mm base profile.
2. Form the R10 rounded corner.
3. Extrude the base to 20 mm.
4. Create the 20 mm diameter hole.
5. Add the rear stepped posts.
6. Sketch a line representing the diagonal support.
7. Use Rib to generate the gusset at the specified thickness.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Inventor Rib command.

MODEL REQUIREMENTS

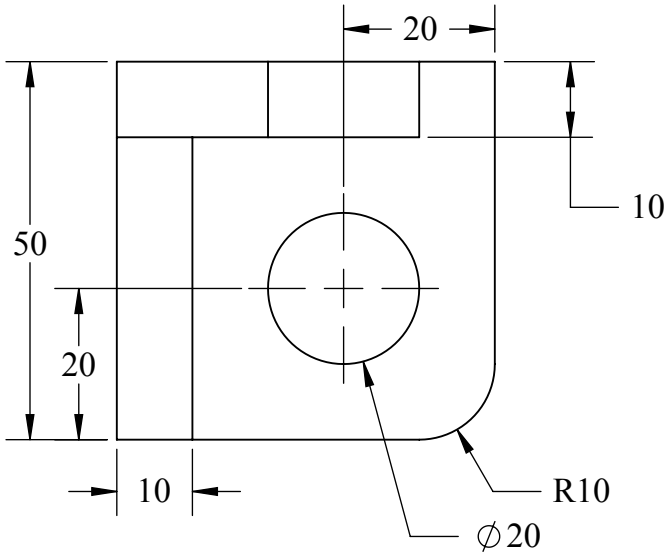
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_26.ipt | Verify Front, Top, Right, Isometric

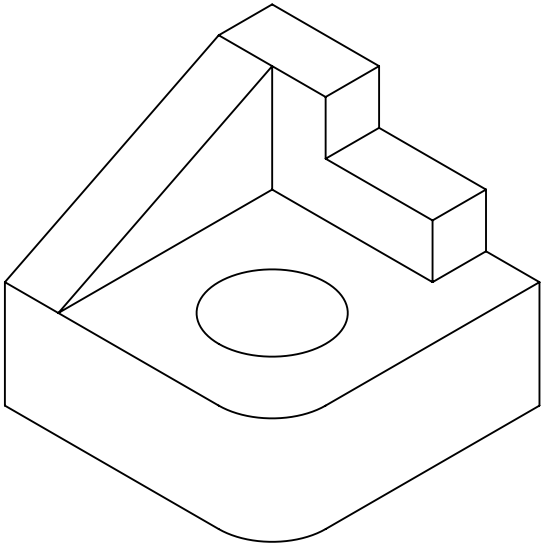
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

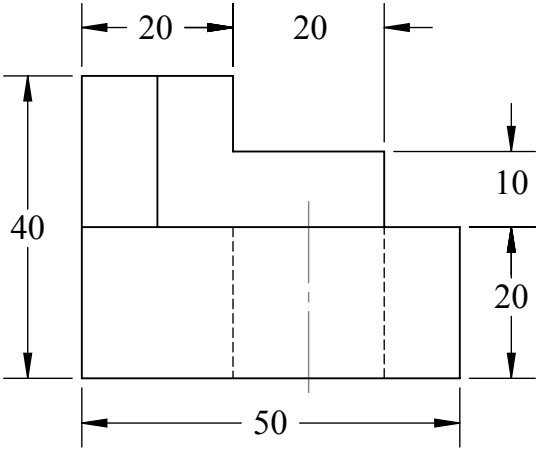
PART 26 | Assigned material: ABS Plastic

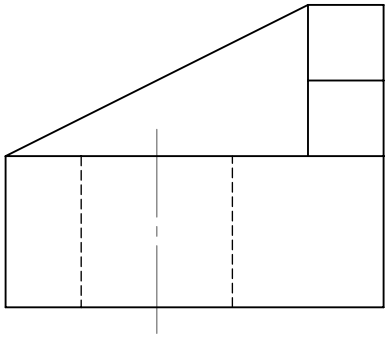
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 26

AUTODESK INVENTOR 2025 - LESSON 27

Rounded Base with Ramp

SUGGESTED COMMANDS

Arc | Extrude | Hole | Line | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the D-shaped base using 40 mm diameter outer geometry.
2. Extrude the base.
3. Add the 20 mm diameter hole.
4. Sketch the rear ramp profile on the side.
5. Extrude the ramp/support across its specified width.
6. Verify the 30 mm and 20 mm height relationships.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

D-shaped profiles and ramp creation.

MODEL REQUIREMENTS

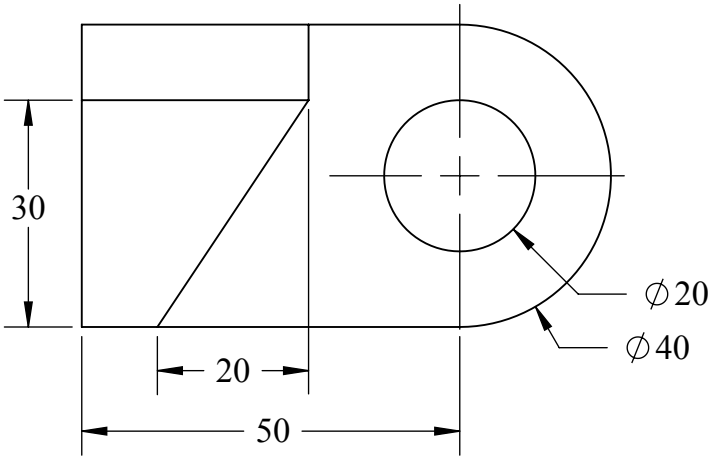
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_27.ipt | Verify Front, Top, Right, Isometric

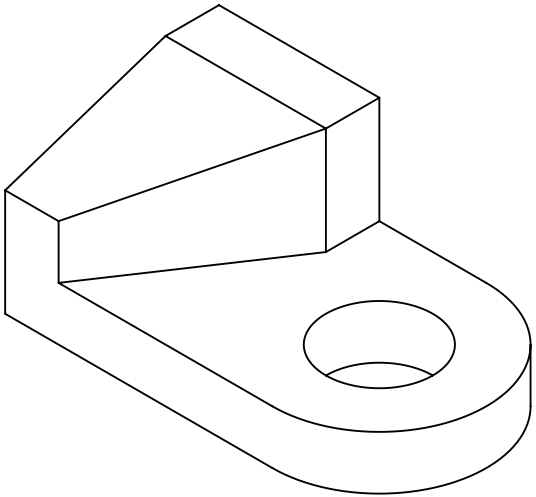
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

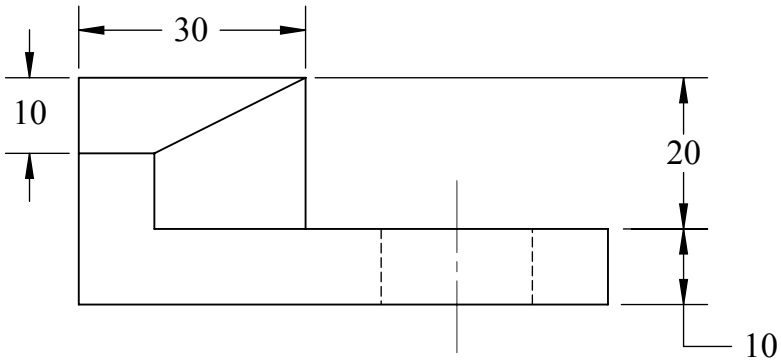
PART 27 | Assigned material: ABS Plastic

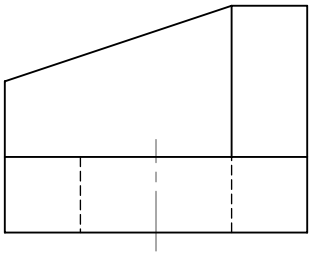
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 27

AUTODESK INVENTOR 2025 - LESSON 28

Upright Saddle and Gusset

SUGGESTED COMMANDS

Extrude | Circle / Arc | Extrude Cut | Rib | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the 70 mm base plate.
2. Add the upright support.
3. Sketch the 20 mm diameter semicircular groove.
4. Extrude Cut the groove through the upright.
5. Add the right-side diagonal gusset with Rib.
6. Create the 45 degree transition/notch.
7. Check all 10 mm thicknesses.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Saddle cuts and gussets.

MODEL REQUIREMENTS

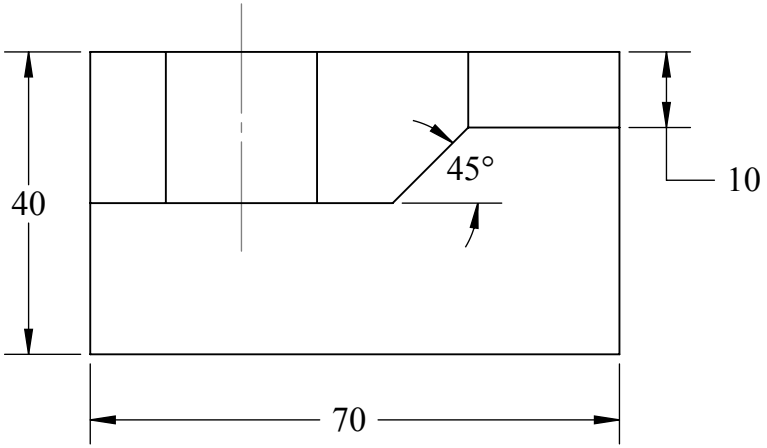
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_28.ipt | Verify Front, Top, Right, Isometric

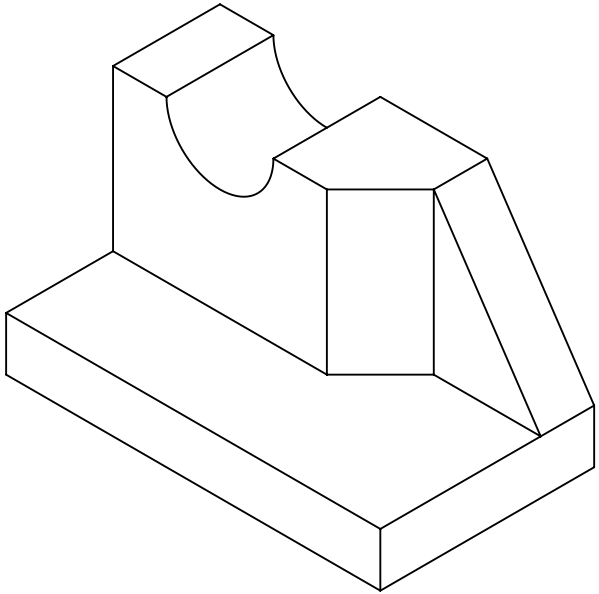
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

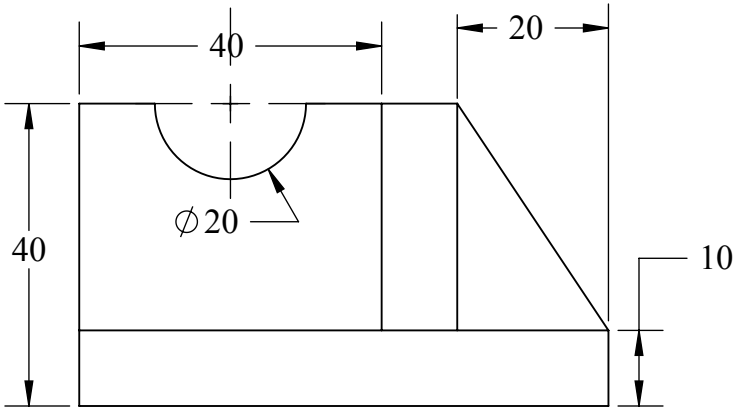
PART 28 | Assigned material: Aluminum 6061

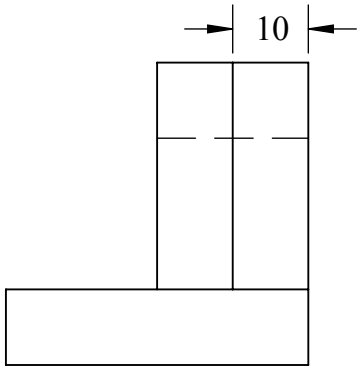
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 28

AUTODESK INVENTOR 2025 - LESSON 29

Opposed Circular Cutouts

SUGGESTED COMMANDS

Circle | Arc | Trim | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the plan profile containing the 40 mm diameter outer arcs.
2. Create the opposing 20 mm diameter semicircular cutouts.
3. Trim excess circular geometry.
4. Extrude the overall profile to full height.
5. Cut the required portion down to the 20 mm level.
6. Verify the resulting stepped S-form.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Circular profile construction and partial-depth cuts.

MODEL REQUIREMENTS

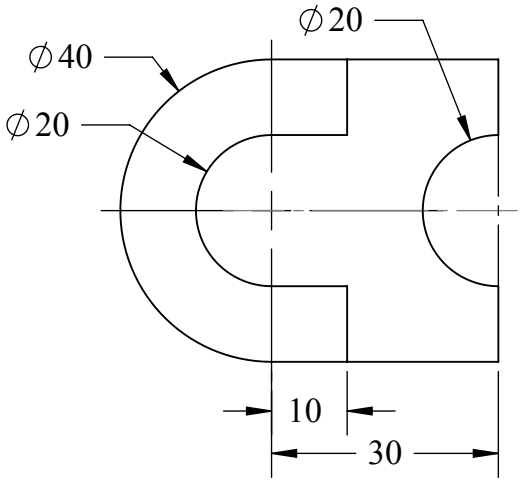
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_29.ipt | Verify Front, Top, Right, Isometric

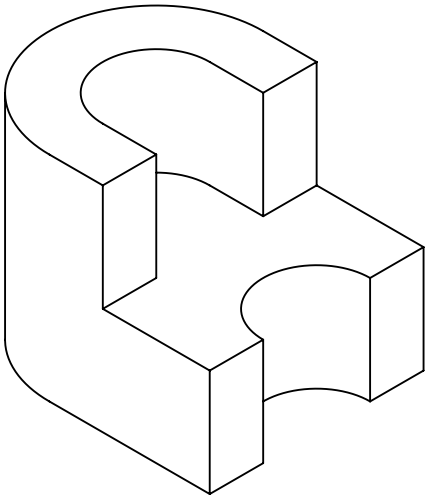
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

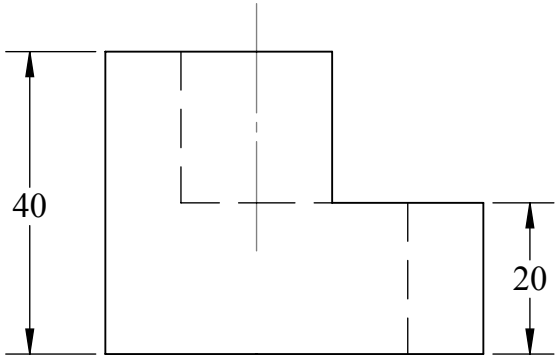
PART 29 | Assigned material: Acetal Resin, White

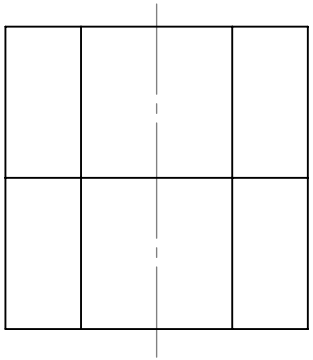
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 29

AUTODESK INVENTOR 2025 - LESSON 30

Corner Mounting Bracket

SUGGESTED COMMANDS

Extrude | Fillet | Hole | Rib / Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the 10 mm base and its curved R10 end.
2. Build the first upright wall.
3. Build the perpendicular upright wall.
4. Add the inclined wall/support.
5. Create each 10 mm diameter hole with Hole.
6. Apply the R10 transition.
7. Verify wall thickness and hole locations from the three views.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Features on perpendicular sketch planes.

MODEL REQUIREMENTS

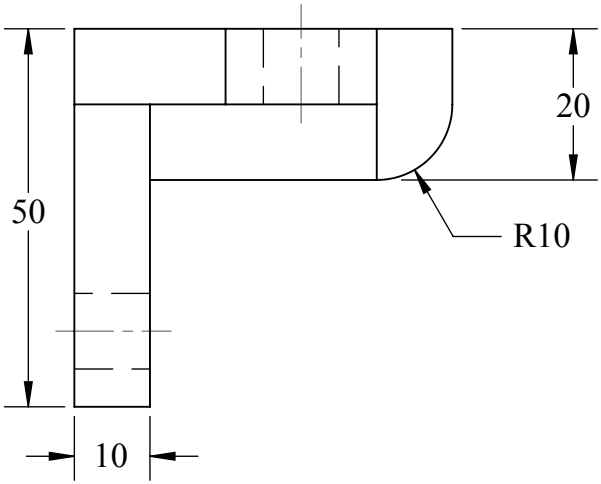
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_30.ipt | Verify Front, Top, Right, Isometric

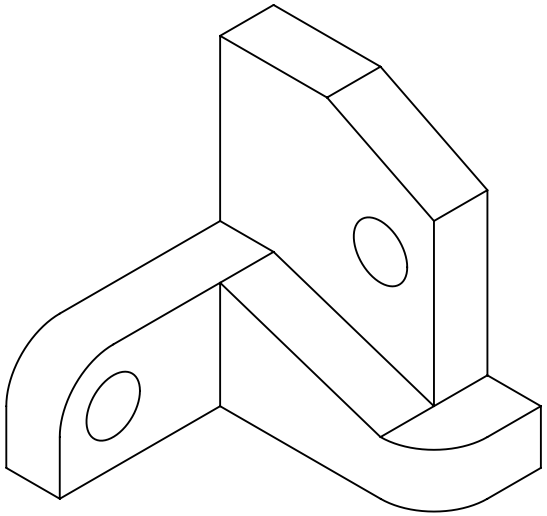
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

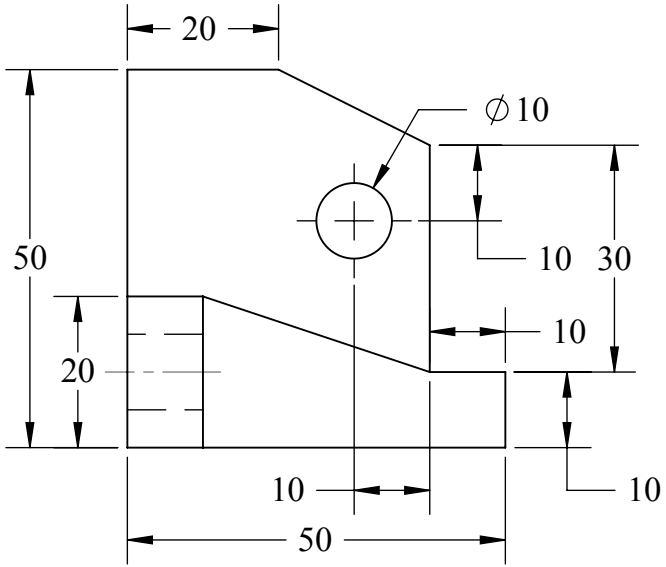
PART 30 | Assigned material: ABS Plastic

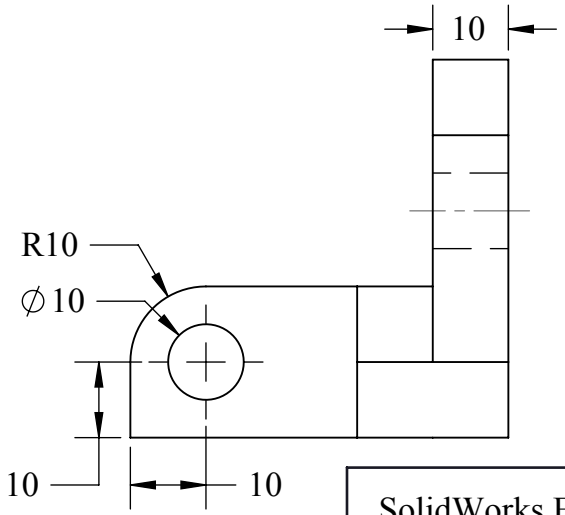
- Verify iProperties: Project (Part Number, Description) and Physical (Material, Mass, Volume).
- Click Update in Physical; submit screenshots/readout. Values depend on actual model geometry.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 30

AUTODESK INVENTOR 2025 - LESSON 31

Ribbed Mounting Base

SUGGESTED COMMANDS

Extrude | Arc | Rib | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the 10 mm bottom plate.
2. Create the R20 rounded end.
3. Add the vertical supports.
4. Form the center U-shaped opening.
5. Add the sloping reinforcement with Rib.
6. Create the 10 mm diameter mounting hole.
7. Verify the 15 mm hole location dimensions.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Ribs and precisely positioned holes.

MODEL REQUIREMENTS

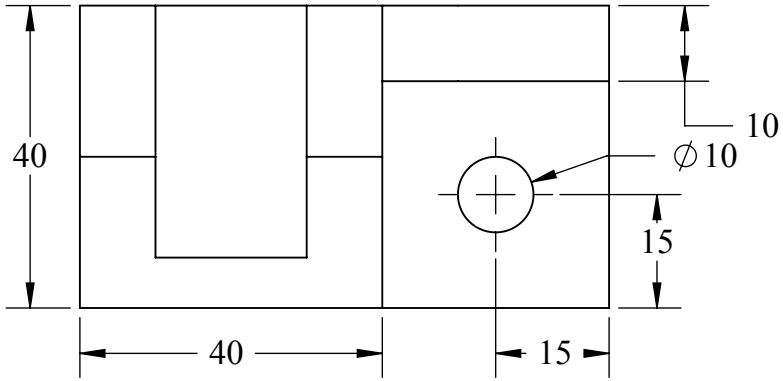
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_31.ipt | Verify Front, Top, Right, Isometric

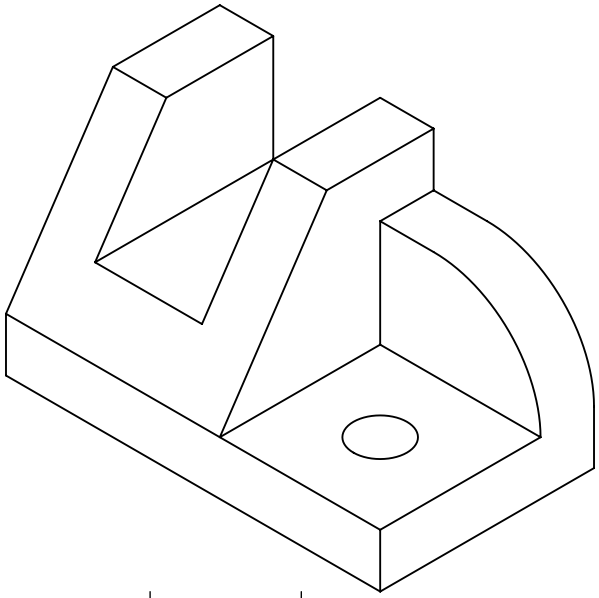
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

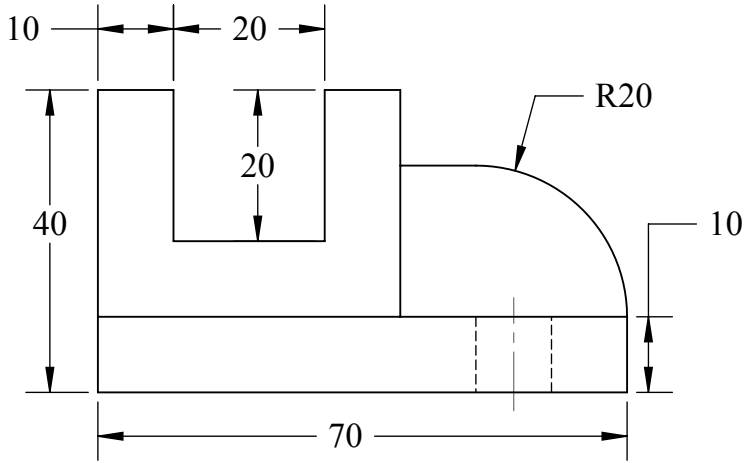
PART 31 | Assigned material: Acetal Resin, White

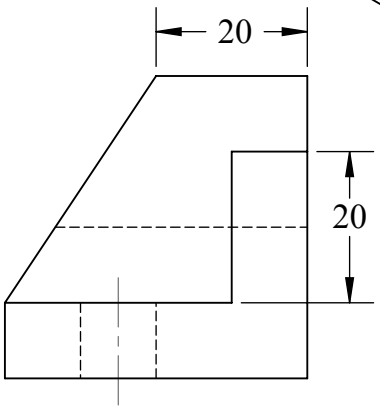
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 31

AUTODESK INVENTOR 2025 - LESSON 32

Keyhole / D-Profile Plate

SUGGESTED COMMANDS

Line | Arc | Circle | Trim | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the entire top profile.
2. Construct the 40 mm diameter rounded outer end.
3. Add the 20 mm diameter concentric hole.
4. Create the two rectangular notches.
5. Trim the sketch to one clean closed profile.
6. Extrude the completed profile 20 mm.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Complex single-sketch extrusion.

MODEL REQUIREMENTS

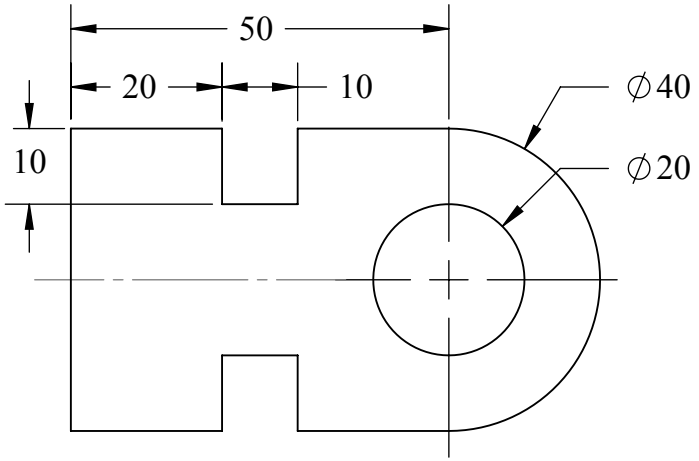
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_32.ipt | Verify Front, Top, Right, Isometric

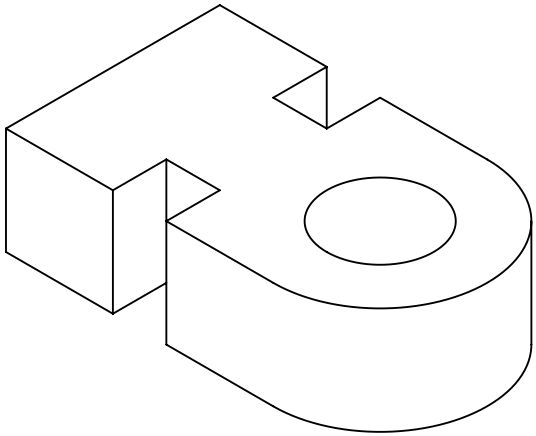
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

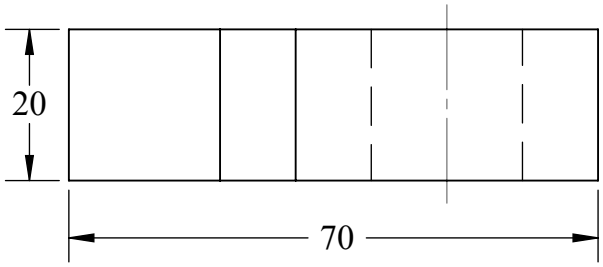
PART 32 | Assigned material: ABS Plastic

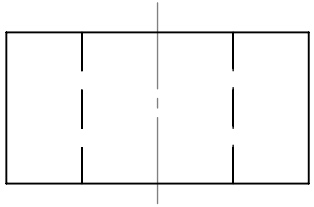
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 32

AUTODESK INVENTOR 2025 - LESSON 33

Seat Form

SUGGESTED COMMANDS

Line | Extrude | Fillet

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the chair/seat side profile.
2. Include the 10 mm base and back thicknesses.
3. Create the inclined seat transition.
4. Extrude the profile to 40 mm width.
5. Apply the four R10 rounds shown.
6. Inspect the resulting seat and back in isometric view.

SUGGESTED COMPLETION TIME

Target time: 10 min Advanced challenge: ~8 min Extended time (2x): 20 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Profile-first modeling and 3D fillets.

MODEL REQUIREMENTS

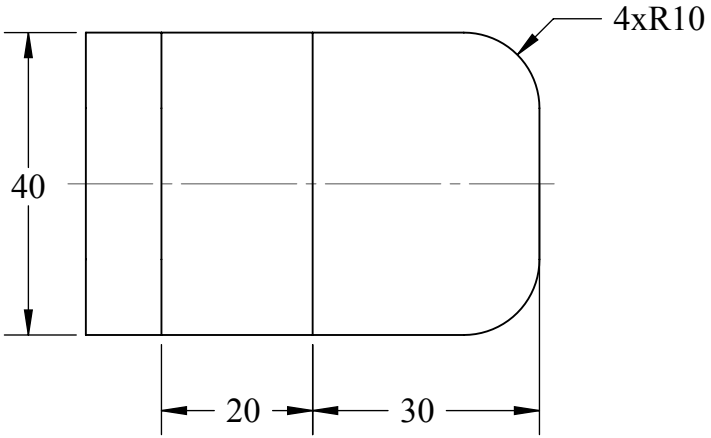
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_33.ipt | Verify Front, Top, Right, Isometric

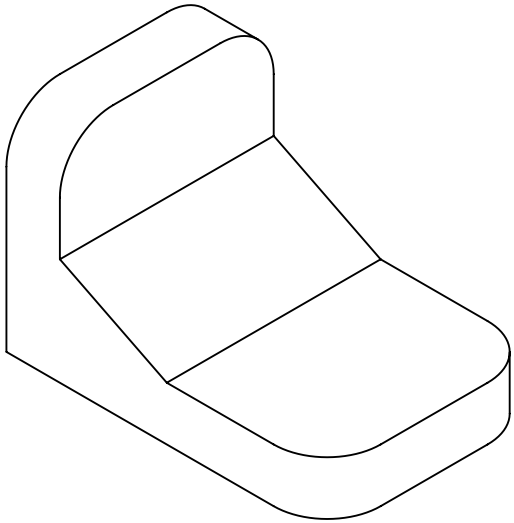
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

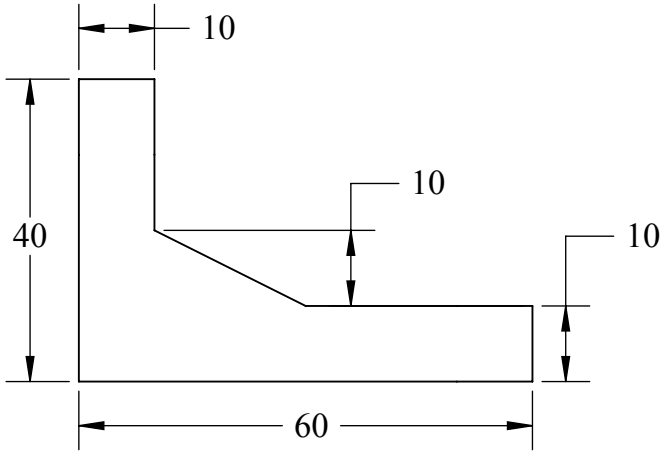
PART 33 | Assigned material: Steel, Mild

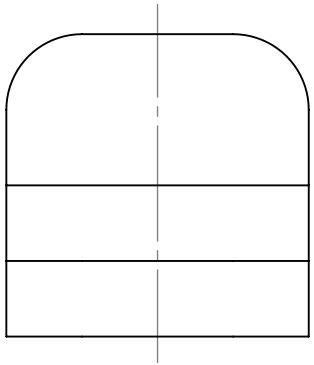
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 33

AUTODESK INVENTOR 2025 - LESSON 34

Stepped Circular Pocket

SUGGESTED COMMANDS

Arc | Extrude | Hole / Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the D-shaped lower body using R20.
2. Extrude the lower level.
3. Add the taller rectangular rear portion.
4. Sketch the 20 mm diameter cylindrical opening/notch.
5. Cut it through the required levels.
6. Verify the 10 and 30 mm step heights.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Cylindrical cuts across stepped solids.

MODEL REQUIREMENTS

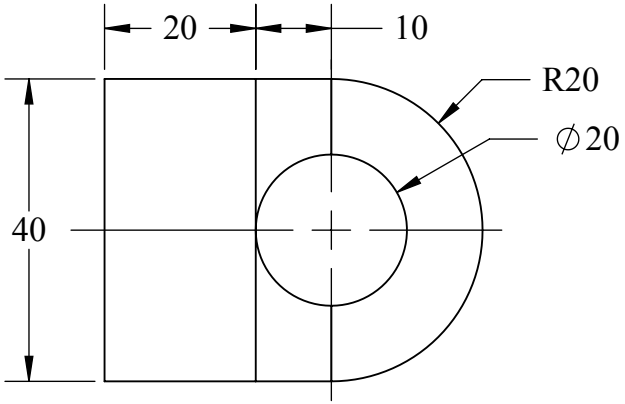
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_34.ipt | Verify Front, Top, Right, Isometric

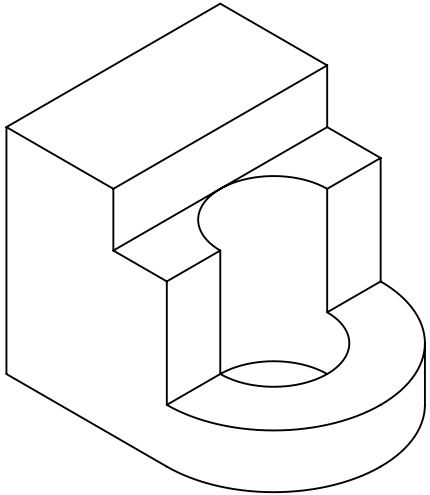
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

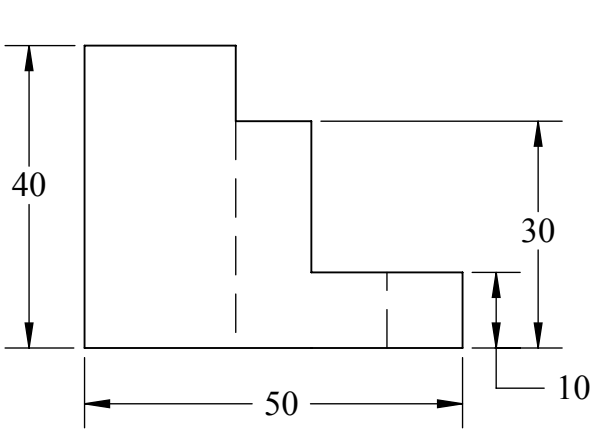
PART 34 | Assigned material: ABS Plastic

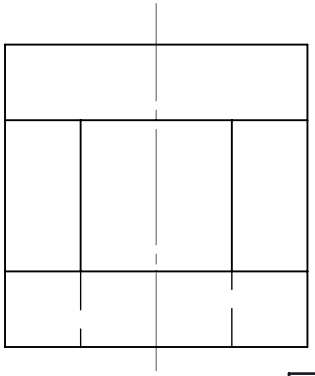
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 34

AUTODESK INVENTOR 2025 - LESSON 35

D-Base with Rear Wall

SUGGESTED COMMANDS

Arc | Circle | Extrude | Hole

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the D-shaped base.
2. Use R20 for the rounded front.
3. Extrude the base 10 mm.
4. Create the 20 mm diameter hole.
5. Add the 30 mm rear wall.
6. Add the smaller 10 mm offset/step.
7. Verify the 50 mm overall length.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Simple multi-level bracket.

MODEL REQUIREMENTS

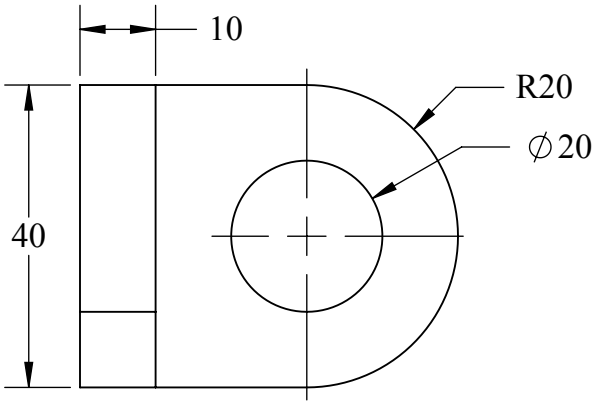
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_35.ipt | Verify Front, Top, Right, Isometric

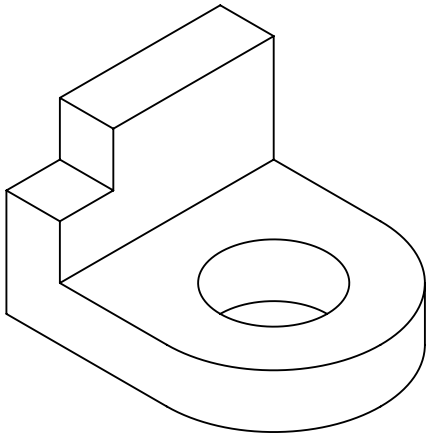
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

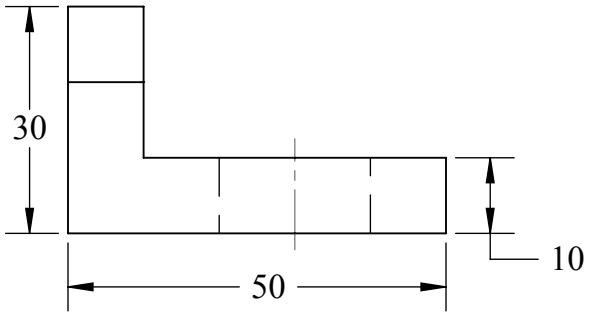
PART 35 | Assigned material: Steel, Mild

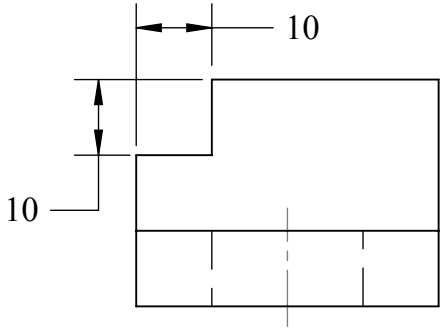
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 35

AUTODESK INVENTOR 2025 - LESSON 36

Curved U-Shaped Block

SUGGESTED COMMANDS

Line | Arc | Sketch Fillet | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the outer boundary.
2. Build the R10 rounded outside transitions.
3. Extrude the full body.
4. Sketch the internal U-shaped opening.
5. Use the specified R10 inner transitions.
6. Cut the center region to the required depth.
7. Verify the final upper/lower levels.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Nested curved profiles.

MODEL REQUIREMENTS

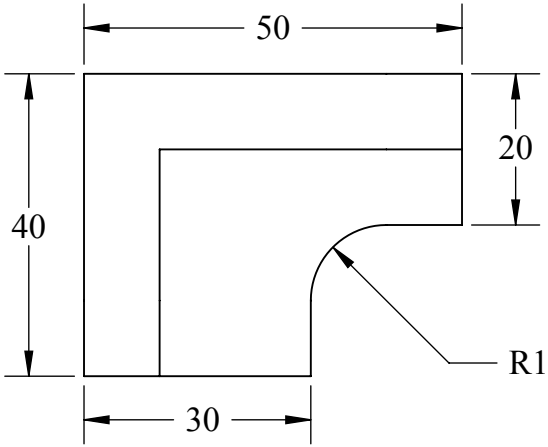
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_36.ipt | Verify Front, Top, Right, Isometric

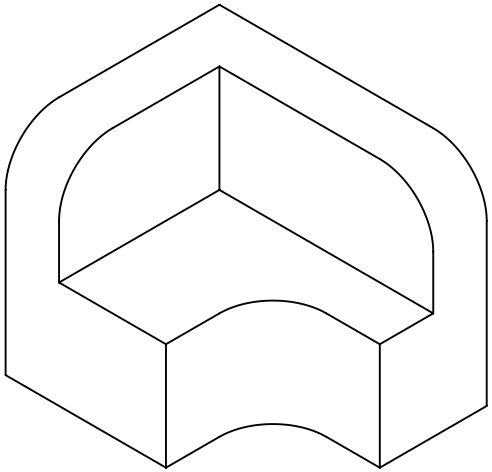
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

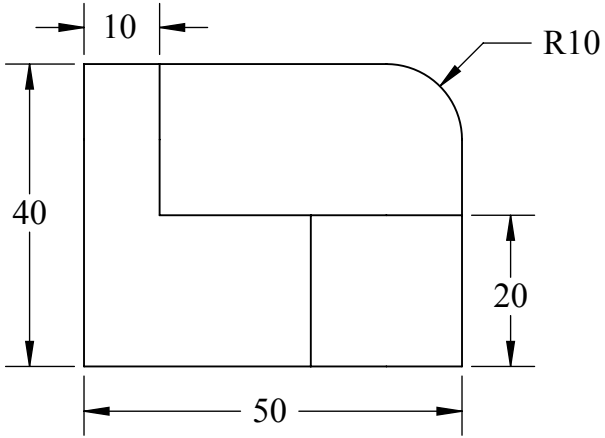
PART 36 | Assigned material: Steel, Mild

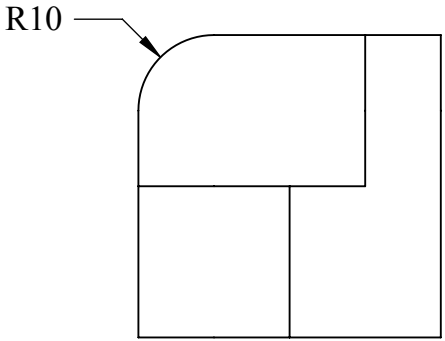
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 36

AUTODESK INVENTOR 2025 - LESSON 37

Raised Ramp with Pocket

SUGGESTED COMMANDS

Arc | Extrude | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the base with R20 rounded front.
2. Extrude it to the lower height.
3. Sketch the raised back/ramp profile.
4. Extrude the raised portion.
5. Sketch the triangular opening on the top.
6. Extrude Cut the triangular pocket.
7. Verify the 10/20 mm heights.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Triangular pockets and rounded bases.

MODEL REQUIREMENTS

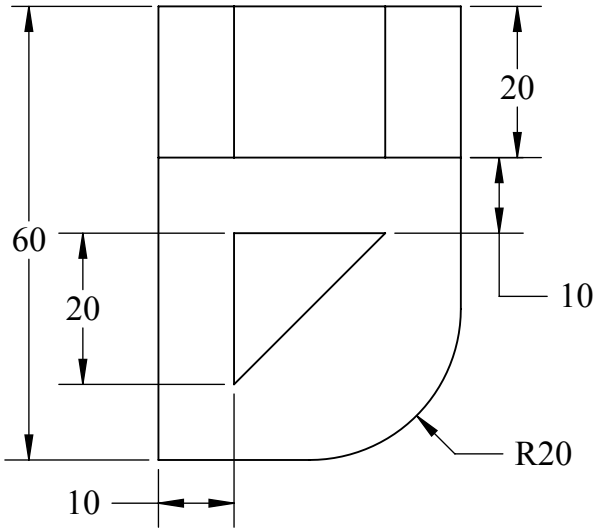
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_37.ipt | Verify Front, Top, Right, Isometric

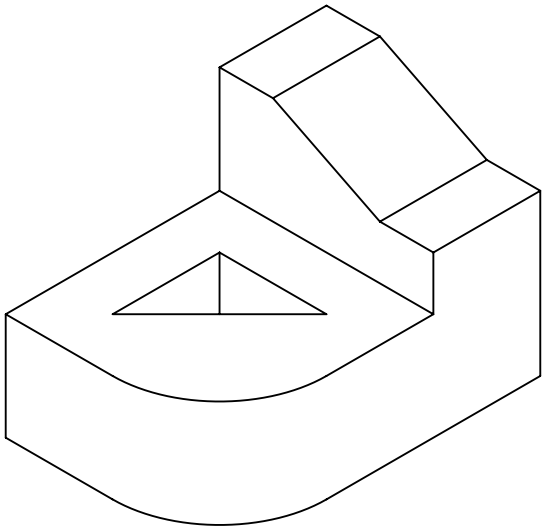
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

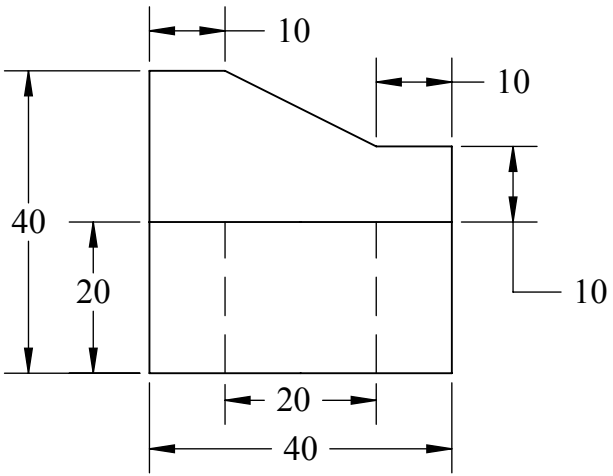
PART 37 | Assigned material: ABS Plastic

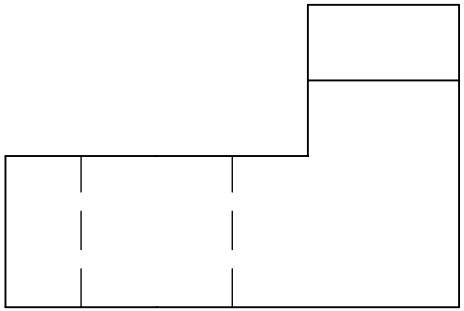
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 37

AUTODESK INVENTOR 2025 - LESSON 38

Semicircular Trough

SUGGESTED COMMANDS

Rectangle | Extrude | Arc | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the main rectangular block.
2. Sketch the R20 semicircular trough on the end face.
3. Extrude Cut the trough along the part.
4. Create the required sloping upper surfaces using triangular cuts.
5. Verify the 40 mm width and 60 mm length.
6. Inspect the trough centerline.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Longitudinal groove creation.

MODEL REQUIREMENTS

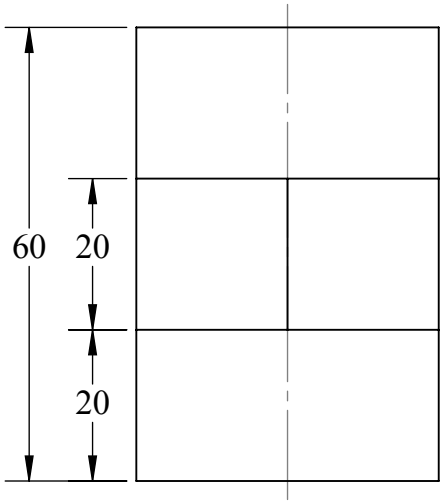
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_38.ipt | Verify Front, Top, Right, Isometric

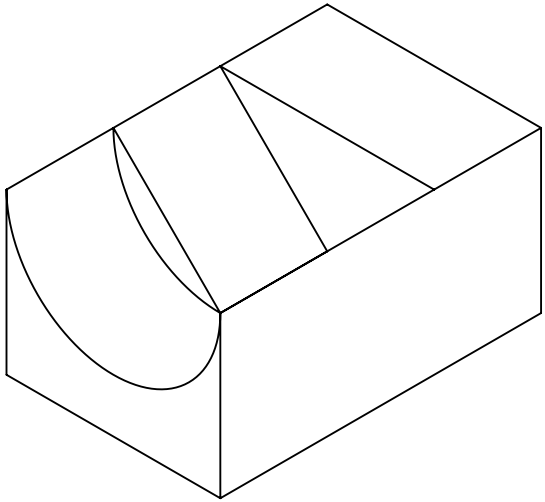
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

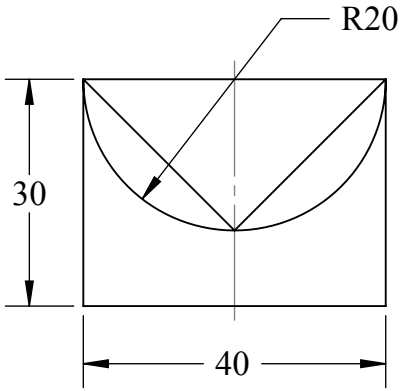
PART 38 | Assigned material: Aluminum 6061

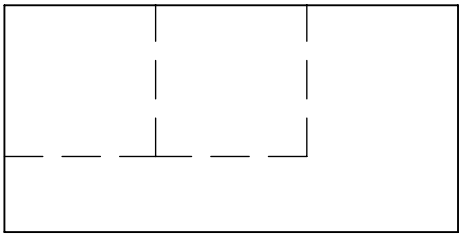
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 38

AUTODESK INVENTOR 2025 - LESSON 39

Quarter-Round Ramp Block

SUGGESTED COMMANDS

Extrude | Arc | Extrude | Line

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the 20 mm-high base.
2. Sketch the upper profile.
3. Construct the R20 curved transition.
4. Extrude the curved raised section.
5. Create the diagonal ramped surface from the side profile.
6. Check the 50 mm overall length and 40 mm height.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Combining curved and inclined surfaces.

MODEL REQUIREMENTS

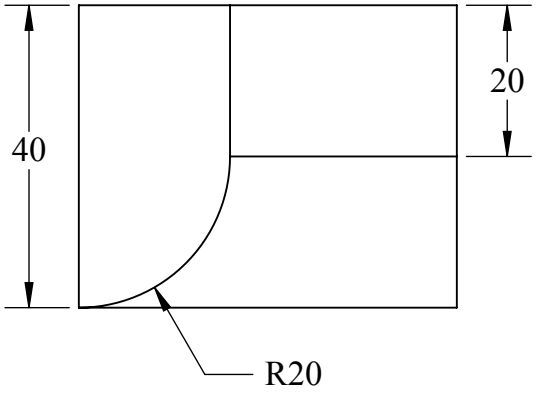
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_39.ipt | Verify Front, Top, Right, Isometric

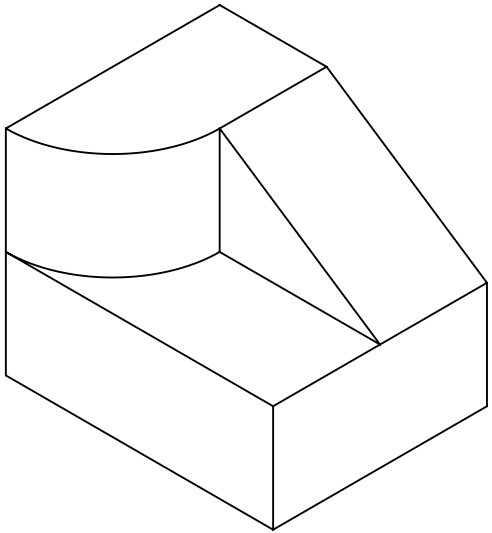
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

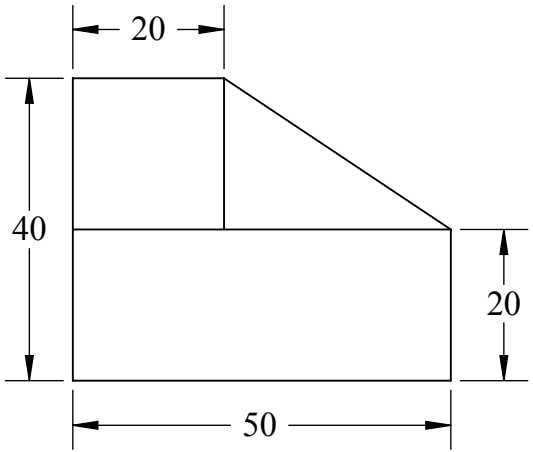
PART 39 | Assigned material: Aluminum 6061

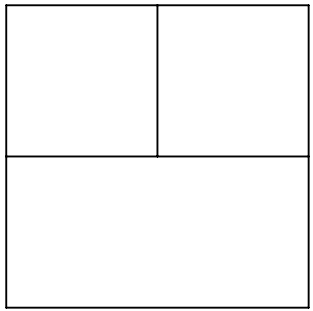
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 39

AUTODESK INVENTOR 2025 - LESSON 40

Faceted Point / Wedge Solid

SUGGESTED COMMANDS

Extrude | Work Plane | Loft | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Begin with the maximum bounding solid.
2. Locate the vertices shown in the three orthographic views.
3. Establish any required Work Plane.
4. Create triangular profiles defining the sloping faces.
5. Use Loft or successive triangular cuts to form the converging surfaces.
6. Verify each diagonal edge against the isometric view.

SUGGESTED COMPLETION TIME

Target time: 22 min Advanced challenge: ~16 min Extended time (2x): 44 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Work geometry and advanced faceted surfaces.

MODEL REQUIREMENTS

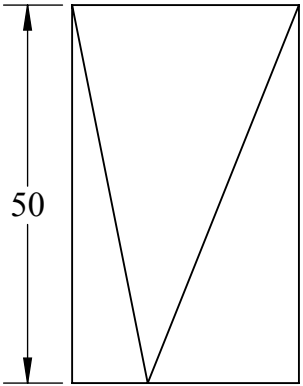
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_40.ipt | Verify Front, Top, Right, Isometric

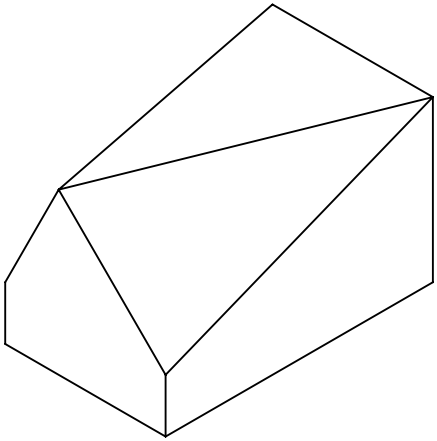
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

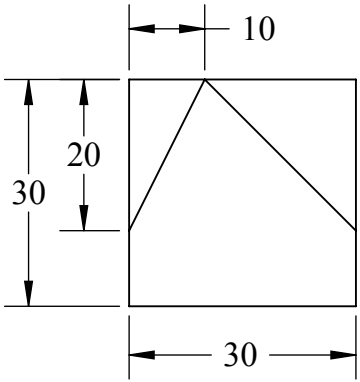
PART 40 | Assigned material: ABS Plastic

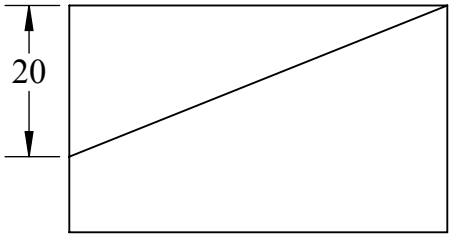
- Check iProperties: Part Number, Description, Material, Mass, Volume; Update Physical.
- Also submit IDW/DWG: Front, Top, Right, Isometric, key dimensions, units and title block.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 40

AUTODESK INVENTOR 2025 - LESSON 41

Inclined Rear Support

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Chamfer / Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the rectangular lower base.
2. Sketch the raised rear support from the side view.
3. Create the 45 degree inclined surface.
4. Extrude the upper support to the required width.
5. Confirm the 60 mm length and 40 mm total height.
6. Compare both side profiles.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Inclined upper features.

MODEL REQUIREMENTS

Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_41.ipt | Verify Front, Top, Right, Isometric

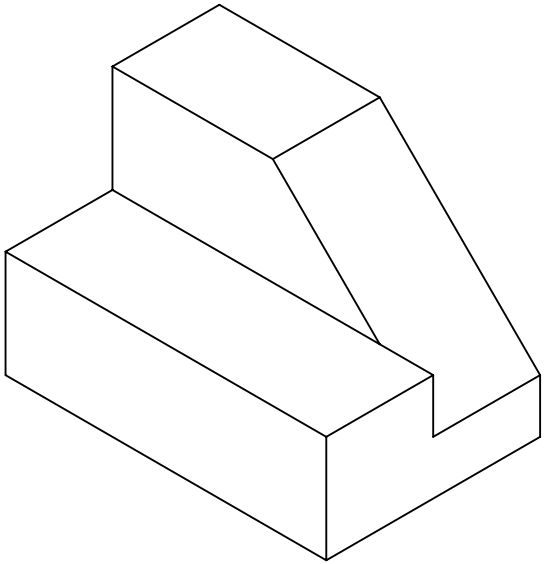
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

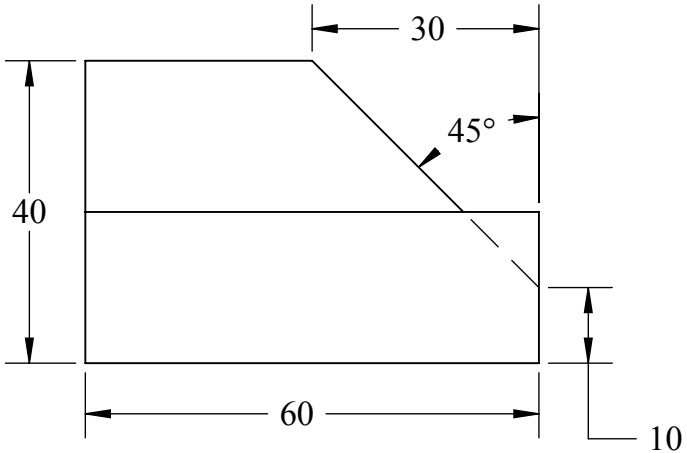
PART 41 | Assigned material: Stainless Steel

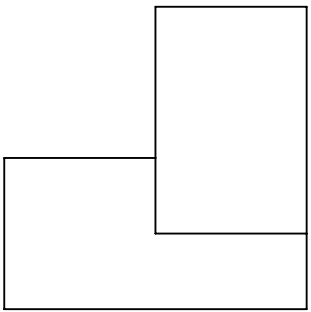
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: NO - inclined solid; no natural cavity.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 41

AUTODESK INVENTOR 2025 - LESSON 42

Twin Upright Block

SUGGESTED COMMANDS

Extrude | Line | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the lower rectangular body.
2. Add the first 10 mm-thick upright.
3. Add the taller rear upright.
4. Sketch the sloping intermediate face.
5. Extrude or cut the slope using the 30/20 mm locations.
6. Verify the 60 x 40 overall footprint/profile.

SUGGESTED COMPLETION TIME

Target time: 10 min Advanced challenge: ~8 min Extended time (2x): 20 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Multiple height levels.

MODEL REQUIREMENTS

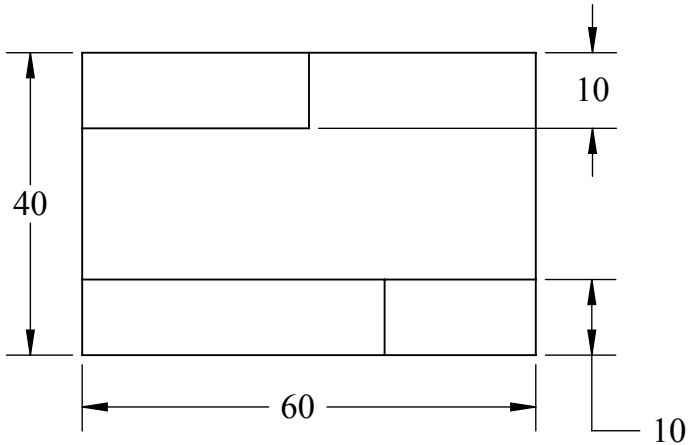
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_42.ipt | Verify Front, Top, Right, Isometric

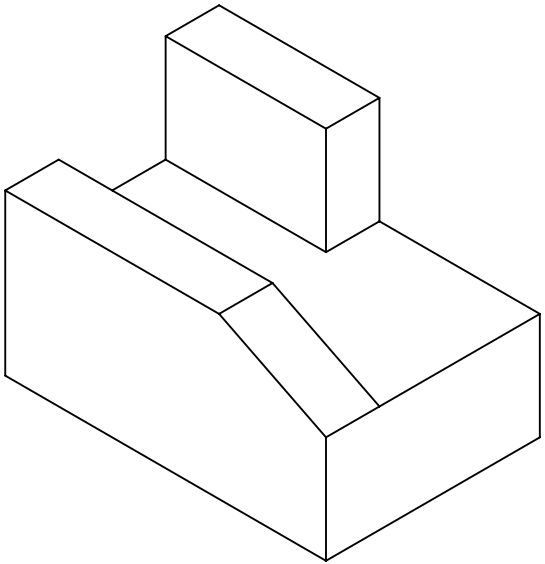
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

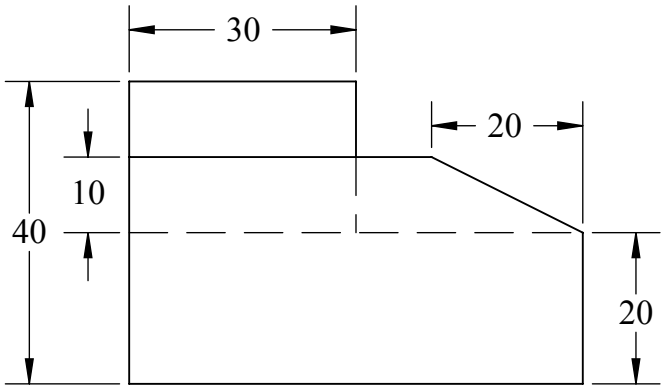
PART 42 | Assigned material: Aluminum 6061

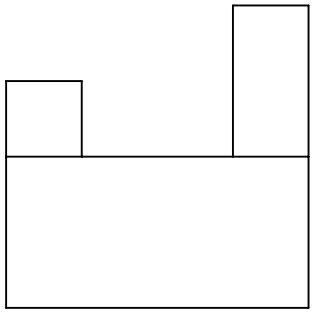
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: OPTIONAL - if an accessible broad face permits hollowing.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 42

AUTODESK INVENTOR 2025 - LESSON 43

Stepped Ramp Assembly

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Build the 10 mm lower base.
2. Add the left stepped portion.
3. Create the 30 mm raised block.
4. Sketch the large diagonal side profile.
5. Extrude the sloping feature to its required width.
6. Verify all 10 and 20 mm offsets.

SUGGESTED COMPLETION TIME

Target time: 12 min Advanced challenge: ~9 min Extended time (2x): 24 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Additive modeling from several profiles.

MODEL REQUIREMENTS

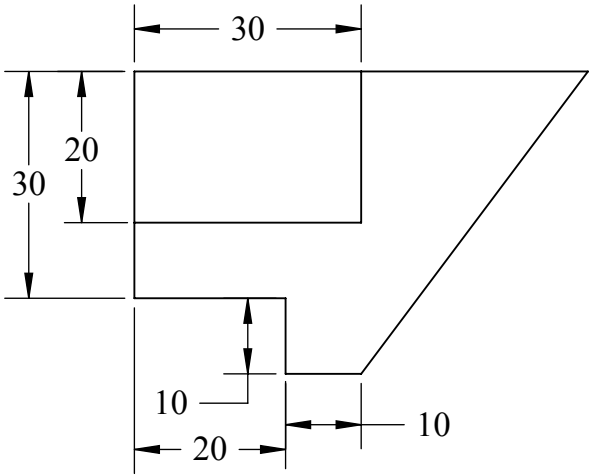
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_43.ipt | Verify Front, Top, Right, Isometric

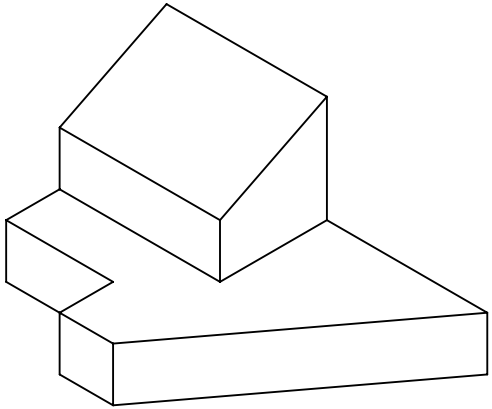
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

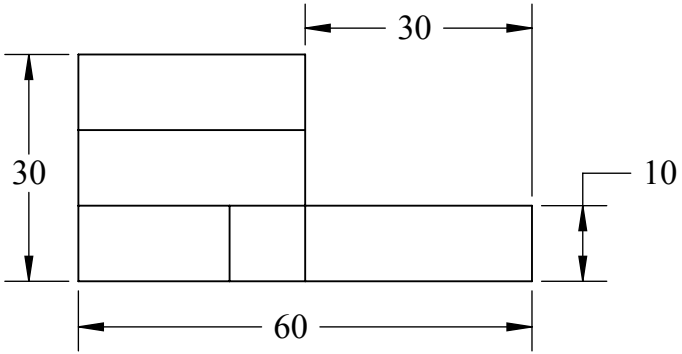
PART 43 | Assigned material: Stainless Steel

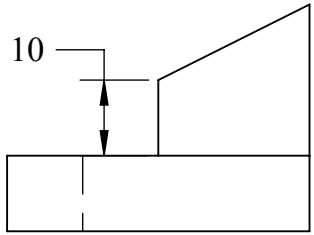
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: NO - stepped geometry must remain exact.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 43

AUTODESK INVENTOR 2025 - LESSON 44

Lobed Base with Inclined Support

SUGGESTED COMMANDS

Circle | Arc | Trim | Extrude | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Construct the lobed base using the 40 mm and 20 mm diameter circular geometry.
2. Use Trim to produce a single closed outline.
3. Extrude the base 10 mm.
4. Locate the upper support.
5. Sketch a single diagonal line/profile.
6. Use Rib or a thin Extrude for the inclined support.
7. Check the 50 mm overall span.

SUGGESTED COMPLETION TIME

Target time: 18 min Advanced challenge: ~14 min Extended time (2x): 36 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Composite circular profiles and ribs.

MODEL REQUIREMENTS

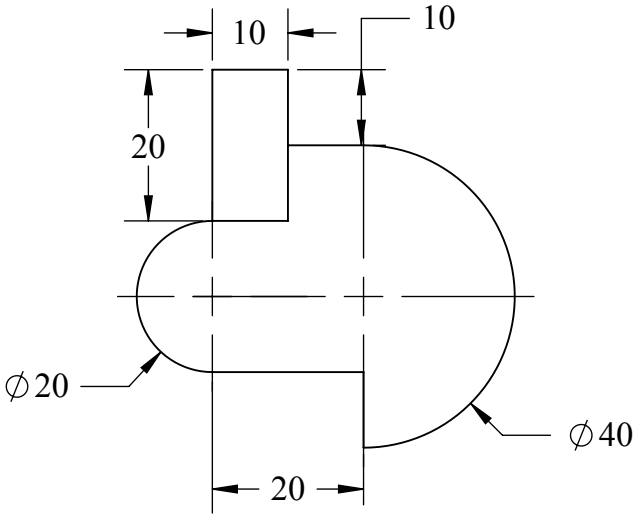
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_44.ipt | Verify Front, Top, Right, Isometric

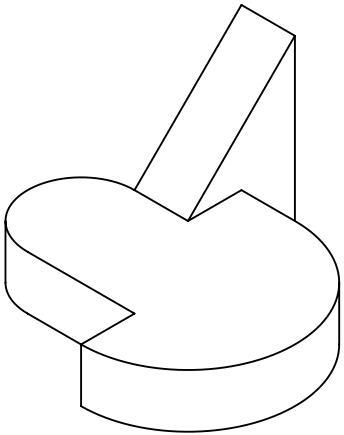
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

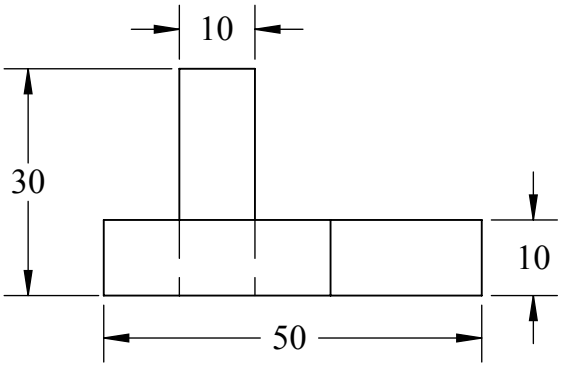
PART 44 | Assigned material: Acetal Resin, White

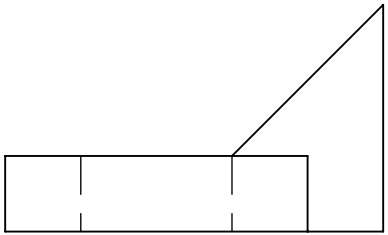
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: OPTIONAL - broad support/base only if wall thickness works.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 44

AUTODESK INVENTOR 2025 - LESSON 45

Curved Frame

SUGGESTED COMMANDS

Line | Arc | Sketch Fillet | Extrude

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the complete side/front frame profile.
2. Construct the outer R20 curve.
3. Add the two R10 inner transitions.
4. Apply Tangent constraints between lines and arcs.
5. Fully dimension the sketch.
6. Extrude the profile to the specified width.
7. Compare the open curved center with the isometric drawing.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Tangent arcs and fully constrained curved profiles.

MODEL REQUIREMENTS

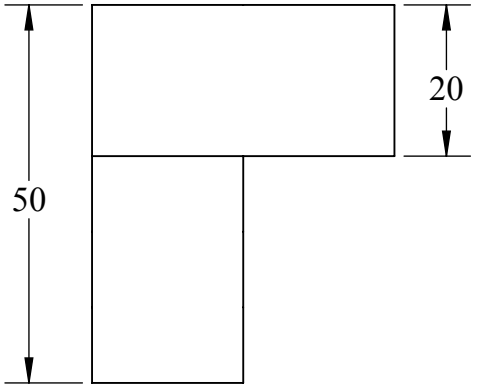
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_45.ipt | Verify Front, Top, Right, Isometric

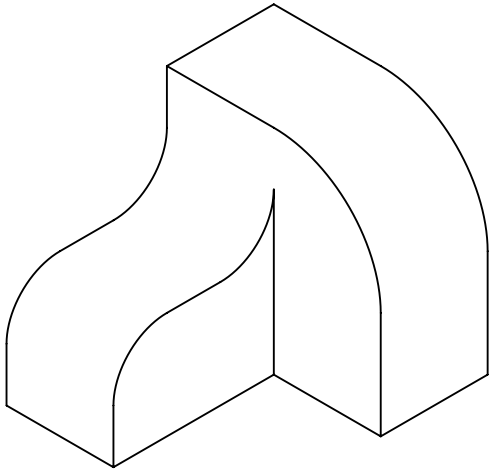
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

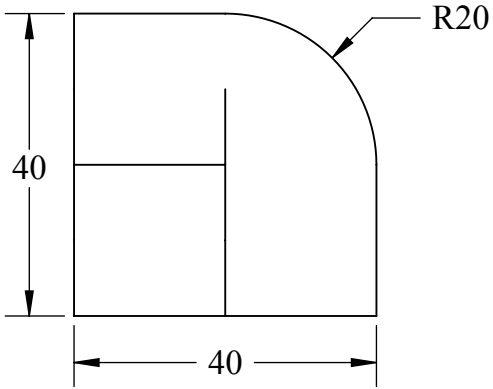
PART 45 | Assigned material: Acetal Resin, White

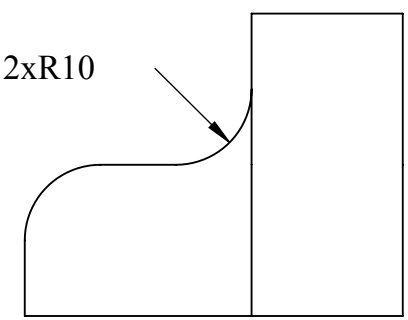
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: YES - curved frame provides suitable cavity.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 45

AUTODESK INVENTOR 2025 - LESSON 46

Stepped Channel Block

SUGGESTED COMMANDS

Rectangle | Extrude | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the 60 x 40 bounding block/body.
2. Establish the 10 mm lower level.
3. Cut the center recess.
4. Create the two raised side portions.
5. Use the 20 mm dimensions to establish slot/recess widths.
6. Check the 30 mm total height.

SUGGESTED COMPLETION TIME

Target time: 10 min Advanced challenge: ~8 min Extended time (2x): 20 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Rectangular pockets and channels.

MODEL REQUIREMENTS

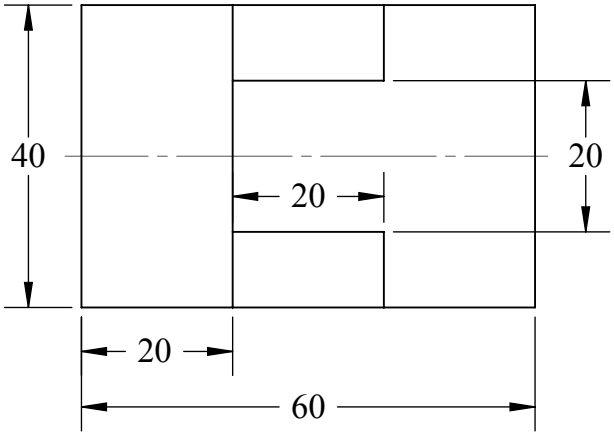
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_46.ipt | Verify Front, Top, Right, Isometric

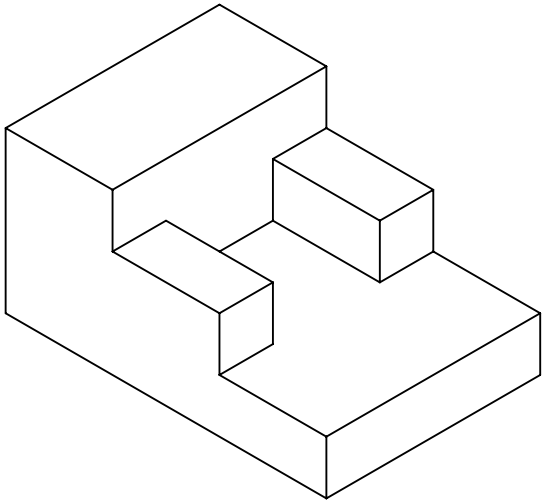
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

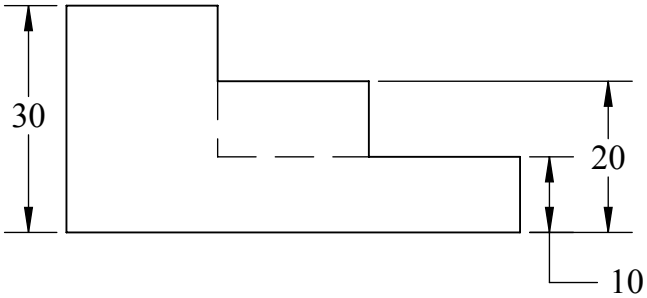
PART 46 | Assigned material: Acetal Resin, Black

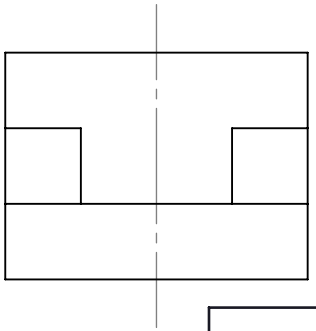
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: YES - channel profile suitable for an open shell.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 46

AUTODESK INVENTOR 2025 - LESSON 47

Rounded C-Block

SUGGESTED COMMANDS

Line | Arc | Extrude | Extrude Cut | Chamfer

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the main stepped side profile.
2. Add the R10 rounded front/lower transition.
3. Extrude the profile to the part width.
4. Cut the rectangular C-shaped opening.
5. Create the 45 degree upper face using Chamfer or an angled cut.
6. Verify the 10/20/30 mm levels.

SUGGESTED COMPLETION TIME

Target time: 15 min Advanced challenge: ~11 min Extended time (2x): 30 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Round-plus-chamfer combination.

MODEL REQUIREMENTS

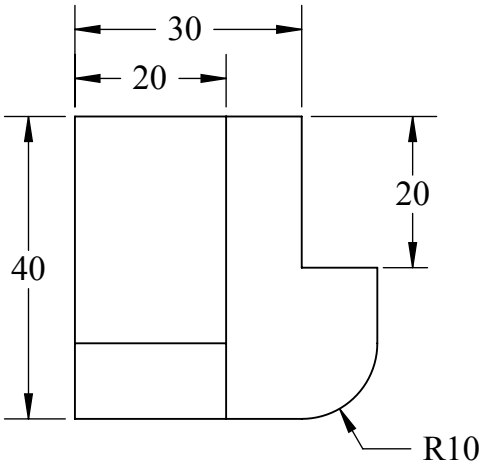
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_47.ipt | Verify Front, Top, Right, Isometric

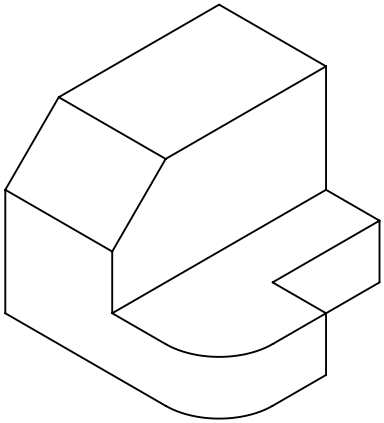
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

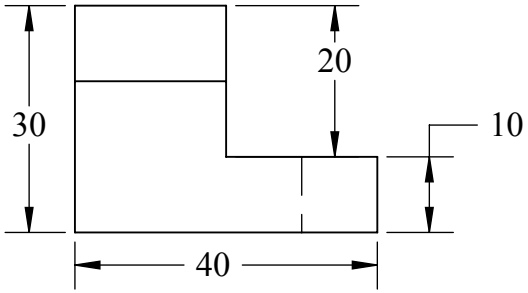
PART 47 | Assigned material: Steel, Mild

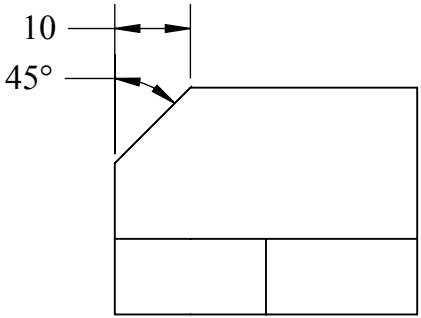
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: YES - C-block can be hollowed from an open face.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 47

AUTODESK INVENTOR 2025 - LESSON 48

Angular Frame with Brace

SUGGESTED COMMANDS

Line | Extrude | Rib

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Sketch the outer C/U-shaped frame.
2. Maintain the 10 mm wall thickness shown.
3. Extrude the main frame.
4. Sketch the diagonal center support.
5. Create the support using Rib or a thin Extrude.
6. Confirm the 40 mm length and 20 mm width/depth relationships.
7. Compare hidden lines with the drawing.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Thin structural brace construction.

MODEL REQUIREMENTS

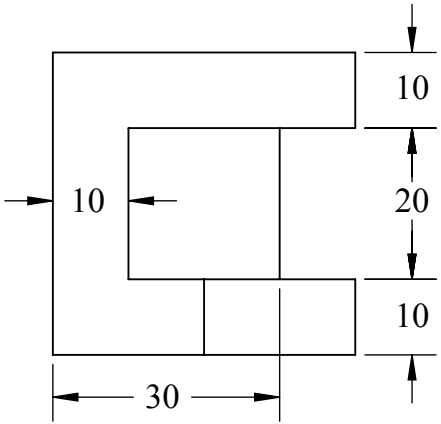
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_48.ipt | Verify Front, Top, Right, Isometric

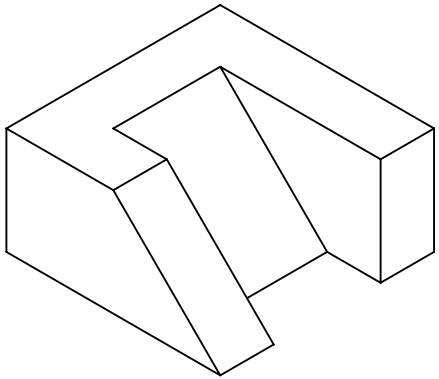
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

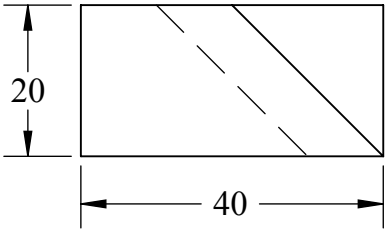
PART 48 | Assigned material: ABS Plastic

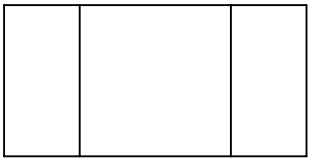
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: OPTIONAL - frame walls, if no feature collapses.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 48

AUTODESK INVENTOR 2025 - LESSON 49

Diagonal Recess Block

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create a 60 x 40 x 40 bounding solid using the drawing dimensions.
2. Sketch the first diagonal boundary on the appropriate face.
3. Extrude Cut the first large recess.
4. Use a perpendicular face to establish the second diagonal.
5. Preserve the 10 mm lower and side wall thickness shown.
6. Inspect the interior inclined face in isometric view.

SUGGESTED COMPLETION TIME

Target time: 14 min Advanced challenge: ~10 min Extended time (2x): 28 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Removing wedges from multiple directions.

MODEL REQUIREMENTS

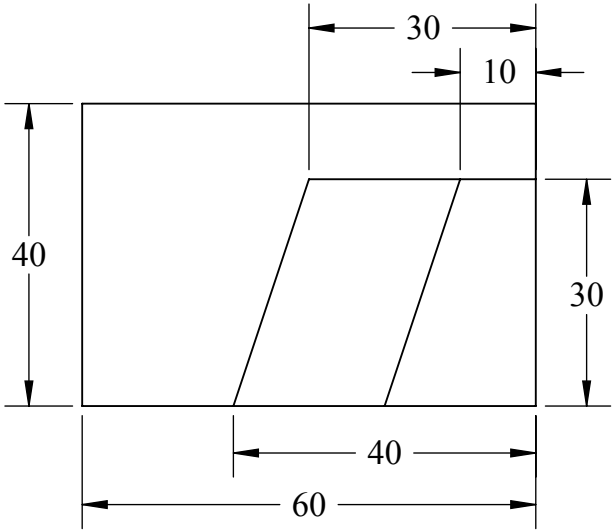
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_49.ipt | Verify Front, Top, Right, Isometric

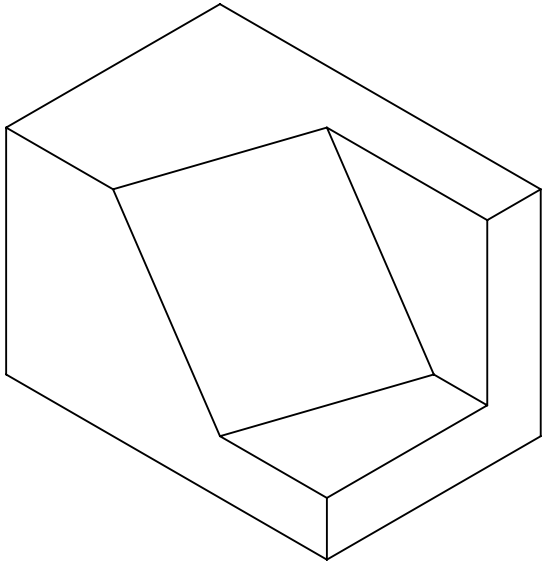
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

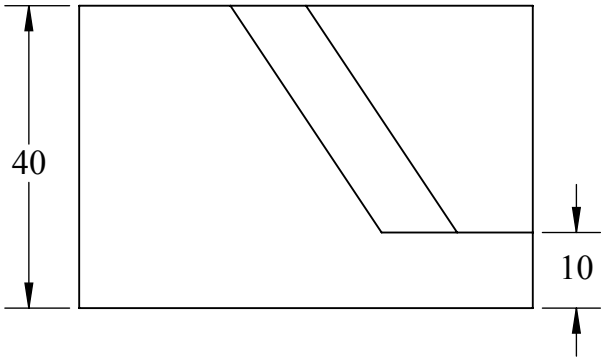
PART 49 | Assigned material: Steel, Mild

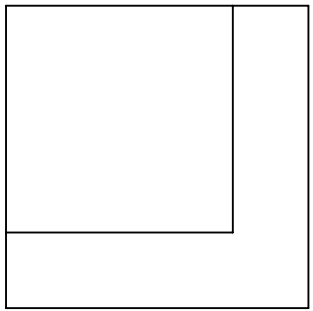
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: NO - diagonal recess is part of required profile.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 49

AUTODESK INVENTOR 2025 - LESSON 50

Intersecting Diagonal Cut Block

SUGGESTED COMMANDS

Rectangle | Extrude | Line | Extrude Cut

OBJECTIVE

Recreate the supplied engineering drawing as a fully constrained Autodesk Inventor part using the drawing dimensions.

STUDENT WORKFLOW

1. Create the maximum 60 x 40 x 40 solid.
2. Sketch the front-view diagonal using the 40 and 30 mm locations.
3. Make the first triangular Extrude Cut.
4. Sketch the second diagonal profile from the perpendicular view.
5. Make the second cut while maintaining the 10 mm offset shown.
6. Verify that the two inclined surfaces intersect at the correct edge.
7. Finish by comparing Front, Top, Right and Isometric views.

SUGGESTED COMPLETION TIME

Target time: 16 min Advanced challenge: ~12 min Extended time (2x): 32 min

Target assumes prior Inventor practice; accuracy and completed geometry remain the priority.

CERTIFICATION / SKILL FOCUS

Orthographic interpretation and intersecting cuts.

MODEL REQUIREMENTS

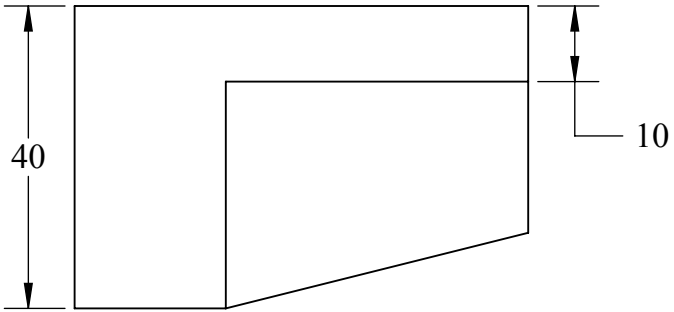
Units: mm | Fully constrain primary sketches | Do not estimate dimensions | Save as SW_Lesson_50.ipt | Verify Front, Top, Right, Isometric

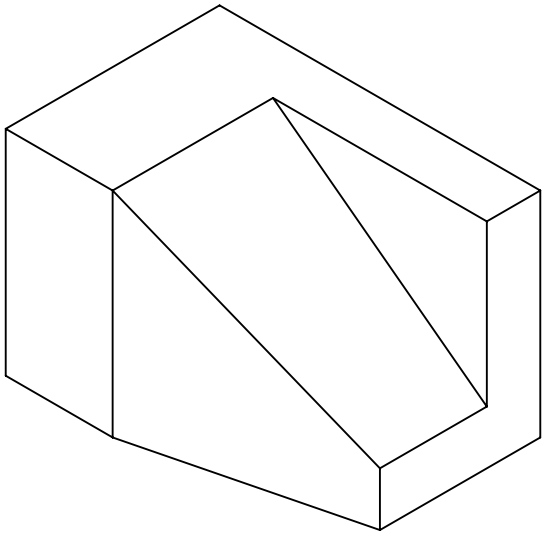
ADDITIONAL INVENTOR 2025 REQUIREMENTS | MATERIALS + VALIDATION

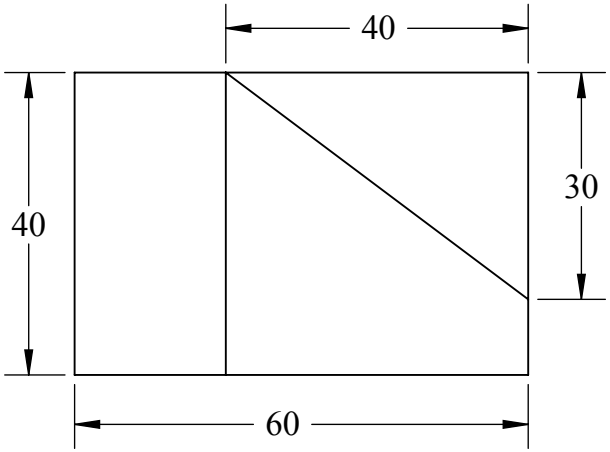
PART 50 | Assigned material: ABS Plastic

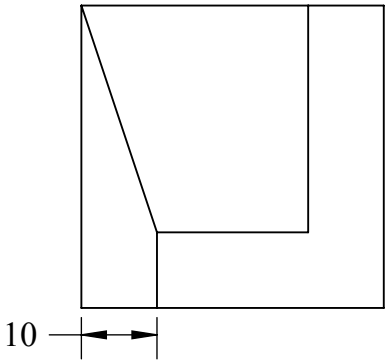
- Check iProperties and submit dimensioned IDW/DWG with 3 views + Isometric + title block.
- Shell evaluation: NO - intersecting cuts are the assessment focus.
- If feasible, use Shell in a separate model state/copy; keep the reference solid unchanged.

Mass is calculated from the finished model and assigned material; do not enter a guessed mass.









SolidWorks Education
Lesson 50

INVENTOR 2025 | REVISED LESSON TIME GUIDE

SW Education Lessons 1-50 | Certification-style pacing

PACING BASIS

Students begin this packet after about 9 weeks / 45 hours of Autodesk Inventor practice.
Target time assumes students already know the basic interface, sketching, dimensions, constraints, and Extrude workflow.
Advanced challenge pace is approximately 75% of the target. Extended-time accommodation is exactly 2x the target.
Times are pacing guides, not automatic grade penalties. Accuracy, geometry, constraints, and file completion remain primary.

LESSON / MODEL	TARGET | 2x
01 Saddle Support	**12 | 24 min**
02 Rounded Base Bracket	**14 | 28 min**
03 Inclined Step Block	**8 | 16 min**
04 Rounded Lug Base	**12 | 24 min**
05 Frame with Curved Corners	**18 | 36 min**
06 Stepped Rounded Block	**10 | 20 min**
07 Base with Circular Support	**15 | 30 min**
08 Arched Support	**15 | 30 min**
09 Twin Inclined Supports	**12 | 24 min**
10 Angled Stepped Block	**10 | 20 min**
11 Multi-Level Block	**12 | 24 min**
12 Angled Wall with Opening	**15 | 30 min**
13 Hipped / Roof Form	**16 | 32 min**
14 Multi-Slope Roof Block	**15 | 30 min**
15 Stair-Step Pocket	**14 | 28 min**
16 Faceted Wedge	**18 | 36 min**
17 Interlocking H Block	**15 | 30 min**

LESSON / MODEL	TARGET | 2x
18 Curved C-Bracket	**16 | 32 min**
19 Internal Stair Frame	**18 | 36 min**
20 Rounded Base with Raised Sup...	**18 | 36 min**
21 Contoured Support	**18 | 36 min**
22 Two-Hole Curved Bracket	**14 | 28 min**
23 Internal Staircase	**12 | 24 min**
24 Inverted Pyramidal / V-Groove ...	**22 | 44 min**
25 Two-Lug Angle Bracket	**16 | 32 min**
26 Base, Post and Gusset	**16 | 32 min**
27 Rounded Base with Ramp	**12 | 24 min**
28 Upright Saddle and Gusset	**16 | 32 min**
29 Opposed Circular Cutouts	**14 | 28 min**
30 Corner Mounting Bracket	**18 | 36 min**
31 Ribbed Mounting Base	**18 | 36 min**
32 Keyhole / D-Profile Plate	**12 | 24 min**
33 Seat Form	**10 | 20 min**
34 Stepped Circular Pocket	**12 | 24 min**

LESSON / MODEL	TARGET | 2x
35 D-Base with Rear Wall	**12 | 24 min**
36 Curved U-Shaped Block	**16 | 32 min**
37 Raised Ramp with Pocket	**14 | 28 min**
38 Semicircular Trough	**12 | 24 min**
39 Quarter-Round Ramp Block	**12 | 24 min**
40 Faceted Point / Wedge Solid	**22 | 44 min**
41 Inclined Rear Support	**12 | 24 min**
42 Twin Upright Block	**10 | 20 min**
43 Stepped Ramp Assembly	**12 | 24 min**
44 Lobed Base with Inclined Support	**18 | 36 min**
45 Curved Frame	**15 | 30 min**
46 Stepped Channel Block	**10 | 20 min**
47 Rounded C-Block	**15 | 30 min**
48 Angular Frame with Brace	**16 | 32 min**
49 Diagonal Recess Block	**14 | 28 min**
50 Intersecting Diagonal Cut Block	**16 | 32 min**

Total target modeling time: 719 minutes (12.0 hours)
Advanced challenge total: ~540 minutes (9.0 hours)

Extended-time total: 1438 minutes (24.0 hours)

Use checkpoints or split complex lessons across class periods when needed.